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Additive manufacturing of magnetic materials

V. Chaudhary[a], S.A. Mantri[c], R.V. Ramanujan[a], R. Banerjee[a,b,c,*]

[a] School of Materials Science and Engineering, Nanyang Technological University, 639798 Singapore, Singapore
[b] Department of Materials Science and Engineering, University of North Texas, Denton, TX 76207, USA
[c] Center for Agile and Adaptive Additive Manufacturing, University of North Texas, Denton, TX 76207, USA

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ABSTRACT

Additive manufacturing (AM) is an attractive process to manufacture net shape, complex, engineering components with minimum waste; however, it has been largely applied to structural materials. AM of functional materials, such as magnetic materials, has received much less attention. Magnetic materials are of high and growing interest in an extremely wide range of applications, e.g., electronic devices, rotating electrical machines, electric vehicles, wind turbines, magnetic cooling, electromagnetic shielding microphones, mobiles, laptops, etc. The processing of functional materials by AM can result in novel magnetic components with improved performance and lower processing cost, motivating the present review on AM of magnetic materials. We review commonly used AM techniques, their working principles, and applications to magnetic materials. We discuss the use of the laser engineering net shaping (LENS) process to produce soft and hard magnets. This technique can also be readily employed to process compositionally graded structures for accelerated materials development through combinatorial/high throughput investigations. Such graded structures can exhibit a wide range of functional and structural properties. The structural and magnetic properties of AM processed Fe-Si, Ni-Fe, Fe-Co, soft magnetic composites, soft magnetic oxides, magnetic shape memory alloys, magnetocaloric alloys as well as high entropy alloys are described. AM of hard magnetic materials, including Alnico, Sm-Co, Nd-Fe-B and Ce-Co alloys is elucidated. The current and future trends in this area are outlined.

1. Introduction

Additive Manufacturing (AM) techniques [1–3] involve printing successive layers of materials from 3D model data in order to fabricate components, usually possessing complex geometries [4–6]. AM has recently transformed from a rapid prototyping technology to a manufacturing technology [7]. However, most of this transformation has been only for structural materials and not for functional materials. AM for functional materials development, e.g., AM to produce soft and hard magnets for electric motors/transformers or other complex components is still in its infancy. Multinational companies, e.g., General Electric (GE), Siemens, HP, Wipro, etc. are developing AM systems and producing complex parts using AM processes. Laser based additive manufacturing is one such promising method for processing components of complex geometry and compositionally graded alloys and components. There are several review papers on the AM of structural materials [4,8–14], however, to date, there is no comprehensive review of AM of magnetic materials.

Metamorphic manufacturing, which relies on local optimization of microstructure [15], requires tight control of process

* Corresponding author.
E-mail addresses: varun004@e.ntu.edu.sg (V. Chaudhary), Raj.Banerjee@unt.edu (R. Banerjee).

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parameters to produce complex geometries. Such components with locally optimized properties can be realized using AM [16]. One objective can be, for example, to process a material which is magnetically soft, mechanically strong, and exhibits high electrical resistivity to fulfil the growing demand for high frequency and high power electric machines operating in extreme conditions. Materials with an optimized combination of mechanical, physical, and functional properties are required in many engineering applications. The traditional method of developing new alloys is to prepare a few samples with different alloy compositions and perform extensive characterization and property assessment. This method is costly and time consuming, limiting the opportunity to discover new excellent materials. Hence, accelerated materials development is of high interest [17–20]. High throughput or combinatorial methods can be used for accelerated discovery of novel materials and to improve the properties of existing materials. These methods require the rapid synthesis of materials over a wide composition range, followed by rapid assessment of the properties. These studies can rapidly reveal compositions with the optimum structure and properties as well as relevant processing parameters. Combinatorial approaches have been extensively used in thin film based materials [21–28]. However, less effort has been devoted to techniques for accelerated development of bulk materials. Diffusion-couple based methods [22,29–33] and additive manufacturing [20,34–38] approaches have been applied to bulk materials to develop materials libraries for combinatorial methodologies.

This paper discusses AM processed magnetic materials, AM process optimization, and potential applications. In the section "Overview of additive manufacturing", commonly used AM techniques for printing magnetic materials, typical AM process parameters and the associated structures are discussed. The section "Soft magnetic materials" provides information on various soft magnetic materials e.g., Fe-Ni, Fe-Si, Fe-Co, soft magnetic ferrites, and soft magnetic composites prepared by AM techniques. The mechanical and magnetic properties of these AM-fabricated magnetic materials are discussed. The AM based combinatorial approach for rapid assessment of bulk magnetic materials is described in the "Graded magnetic materials" section. The change in properties as a function of the process parameters of AM and composition of graded high entropy alloys and Fe-Si alloys are discussed. The structure, magnetic properties and the effect of post processing parameters on the properties of AM processed magnetic shape memory alloys are delineated in the "Shape memory alloys" section. In the section "Magnetocaloric heat exchanger" La-Si based magnetocaloric heat exchangers and their magnetocaloric properties are presented. The magnetic properties of Nd-Fe-B (bonded, hybrid and sintered), Alnico, SmCo and Co-Ce-Fe-Cu permanent magnets prepared by AM techniques are described in the section entitled "Hard magnetic materials". Prototypes of magnetic cores and rotors are discussed in "Prototyping of additive manufactured magnetic materials". Challenges in printing magnetic materials and possible solutions, including the use of artificial intelligence (AI) are described in "Challenges associated with AM prepared magnetic materials". This review paper ends with a summary and outlook section.

2. Overview of additive manufacturing techniques

The American Standards of Testing of Materials (ASTM) defines AM as a process of joining materials to make objects from 3D model data, generally layer upon layer, as opposed to subtractive manufacturing methodologies [39]. While this definition is broadly applicable to metals, polymers, ceramics, composites, and even biological systems, the present overview will focus on additive manufacturing techniques for metallic systems. Bourell et al reported 3D printing of metals and alloys (Cu, Sn, Pb–Sn solder) *via* a selective laser sintering (SLS) process [40] in 1990 [41] The shift from the term SLS to selective laser melting (SLM) [42–45] became widely accepted since the metal powders have to be melted and not just sintered.

2.1. Classification of metal additive manufacturing

The ASTM F42 committee (Additive Manufacturing) classifies metal additive manufacturing techniques into four different classes: (i) Powder bed fusion (PBF), (ii) Direct energy deposition, (iii) Binder jetting, (iv) Sheet lamination [39]. These techniques are further classified depending on the characteristics of the heat source and the feedstock form.

2.1.1. Powder bed fusion

Processes in which a focused energy beam is used to melt a layer of powder on a bed fall under the category of powder bed fusion. Depending on the source of the energy, they are classified as either laser beam (SLM) or electron beam (EBM) processes. The first step in an AM process is to use a computer aided design (CAD) file to create a stereolithography (STL) file for the layer by layer deposition. The re-melting of the $(n-1)^{th}$ layer during the melting of the nth layer results in suitable bonding of the layers. For both laser-based and electron-beam processes, processing parameters such as scan speed, power, scan strategy, etc., play a major role in the final condition of the build [46].

Different manufacturers have adopted various terminologies for metal based powder printing processes. The term selective laser melting (SLM) is used for a metal based powder fusion bed process which uses laser as its heat source. SLM Solutions, Renishaw, EOS GmbH, Concept Laser, etc. are some of the companies which produce laser-based powder bed fusion equipment. Fig. 1 is a schematic of a SLM machine [47]. A system of lenses and mirrors is used for manipulating the position of the laser beam on the powder bed. For the powder delivery system, SLM systems use a soft distribution coater blade to drag the powder across the build surface. These coaters are usually made of rubber or soft carbon fiber bristles. The powder is stored in a supply platform or a feeding station. The coaters are used to rake the powder from the supply platform across the bed onto a substrate plate on which deposition takes place. Once the first layer is deposited, the build platform on which the substrate is located lowers and a new coat of powder is laid. This process is repeated until a 3D component is built. Shielding gases, such as argon, are used to maintain an inert atmosphere during deposition, especially for reactive elements like magnetic alloys. Nitrogen can also be used to create *in situ* reactions between the material and the gases [48,49]. The flow rate of these shielding gases also plays a role in the values of porosity, temperature profile

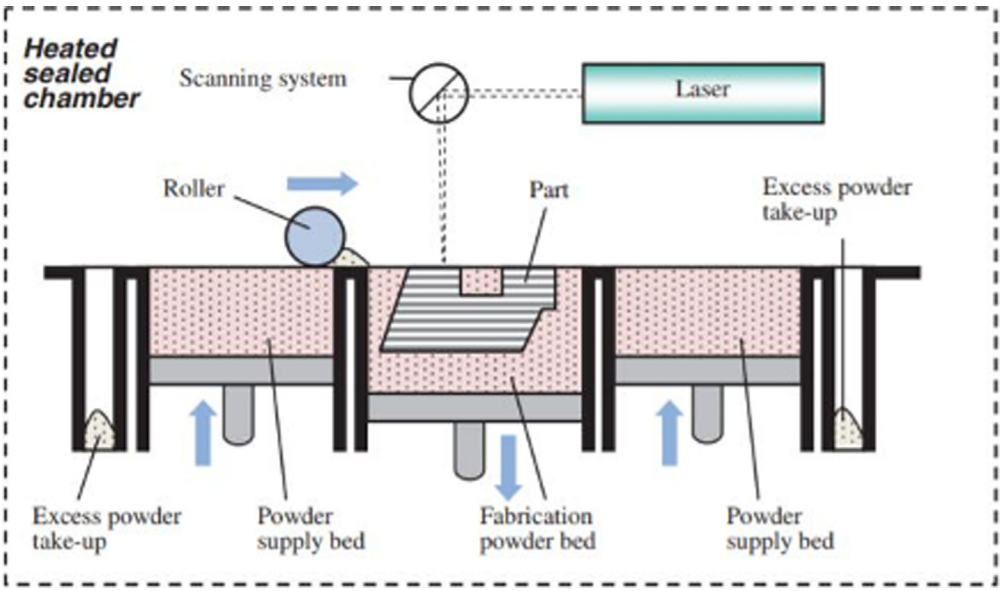


Fig. 1. Schematic of Powder Bed Fusion Process. Reprinted with permission from [47] © 2017, World Scientific Publishing.

and local oxidation [4,46].

Electron beam melting (EBM) [50] is the term for the electron energy based powder bed fusion process. Arcam AB EBM is the most widely used EBM system. An EBM system contains a filament and an electron beam column. Unlike the SLM, an EBM uses magnetic coils to collimate and deflect the electron beam. A schematic (courtesy of Arcam AB) is shown in Fig. 2 [51].

The powder delivery system is slightly different compared to the SLM process since a metal rake is used to coat the powders onto the build platform. The powders surrounding the plate are sintered to provide more stability to the build plate, which is stationary during the recoating process. Since EBM requires electrons for heating, the process takes place under vacuum, usually at pressure of less than 5×10^{-2} Pa (chamber pressure). Major differences in structure and properties are noted due to changes in chamber pressure. This will be further discussed in the relevant sections.

2.1.2. Direct energy deposition

As the name suggests, in the direct energy deposition (DED) process, a feedstock of either powder or wire is deposited onto a melt pool which is generated *via* a laser or electron beam. The most commonly used DED technique is Laser Engineered Net Shaping (LENS™), the relevant equipment is manufactured, e.g., by Optomec Inc. A direct metal laser sintering (DMLS) system is manufactured by EOS. A schematic of these processes is shown in Fig. 3 [52].

As in the case of PBF systems, the DED process requires a powder feeder, a build platform with a substrate plate, and a system of lens and mirrors to manipulate the laser beam. Unlike PBF systems in which the powder is laid over a bed, the powders are transported by inert gases *via* nozzles, onto the build platform. A melt pool is generated *via* the laser beam, the powder flowing through the nozzles deposits on the substrate or on the previously deposited layer. This process is repeated to fabricate the final 3D component. DEDs are also classified based on whether the deposition head moves while the work piece is stationary or, alternatively, the deposition head is stationary and the work piece moves. This feature results in DED techniques having a major advantage over PBF systems since they can fabricate large build volumes and refurbish damaged or worn parts. Wire-fed DED processes (Fig. 4), e.g., EBAM from Siyaki, are extensions of welding techniques. The major advantages of wire-fed DED processes are the speed at which deposition occurs and the capability to make much larger components compared to other DED or PBF processes. However, the resulting material may require more finishing compared to other processes.

2.1.3. Binder jetting

Binder jetting is essentially a powder metallurgy process. The technology is often referred to as 3DP. ExOne is currently the main manufacturer of Binder Jetting printers. As with other powder-based methods, the object being printed is self-supported within the powder bed and is removed from the unbound powder once completed [53]. Two materials are used in the process of binder jetting: a powder and a binder. The binder glues the layers. While the build material is in the powder form, the binder is usually in liquid form. The deposition head alternatively deposits layers of the build material and the binding material while moving along the x and y axes [54]. A schematic of binder jet process is shown in Fig. 5 [55].

The object being printed is lowered on its build platform after each layer is deposited, as in the case of the PBF process. After the build is completed, curing is performed to increase mechanical strength. After curing, the part is usually heat treated for 24–36 h to sinter the loose powder. This also helps to burn off binder, leaving a 60% dense sintered metal part. Hence, the mechanical properties are not always suitable for structural applications. Despite the relatively high speed of printing, additional post processing can add

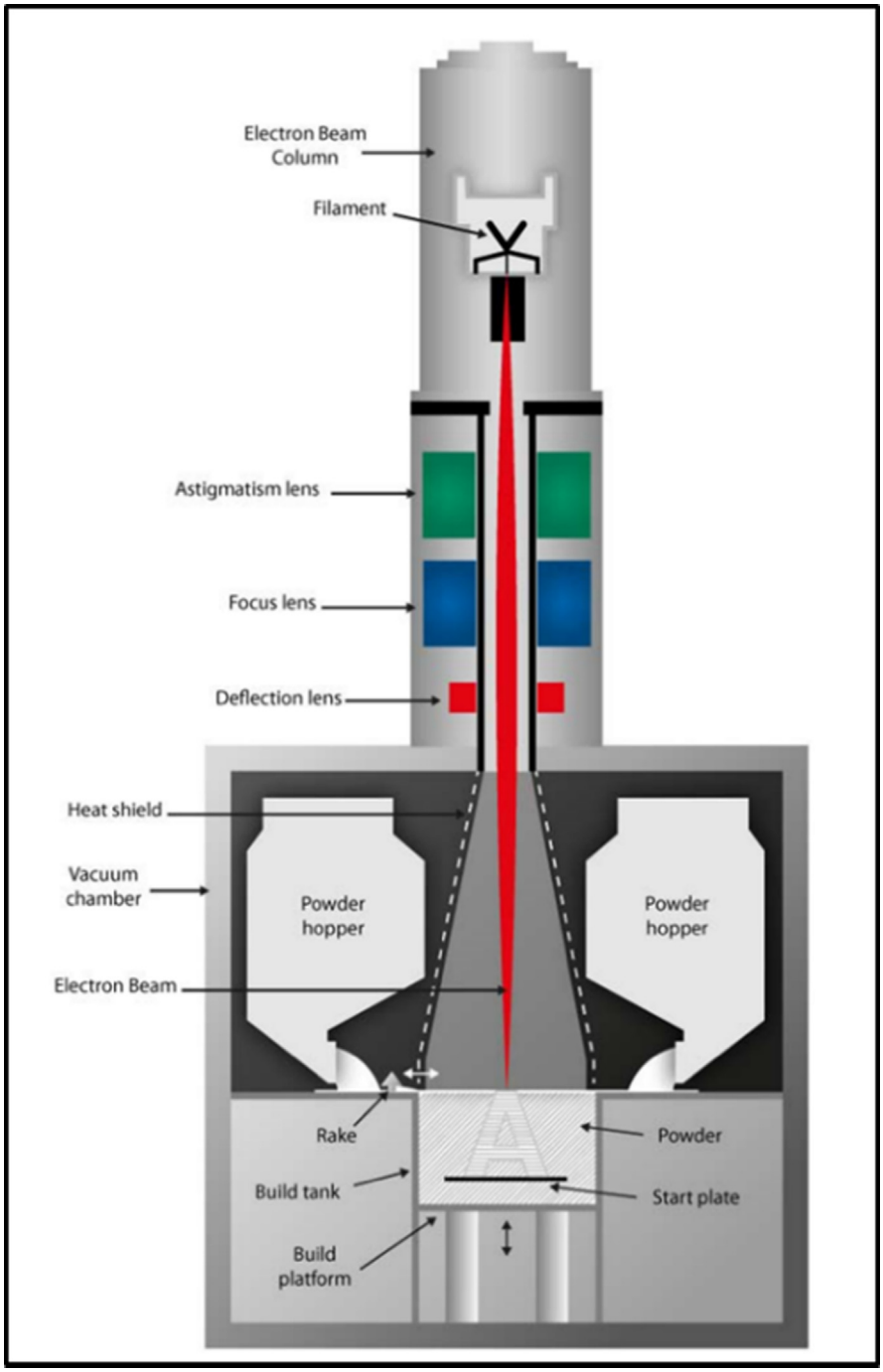


Fig. 2. Schematic of an electron beam melting (EBM) setup. Reprinted from [51] © Arcam AB.

significant time to the process.

Inkjet printing is another 3D printing method which has been useful in prototyping magnetic materials [56–59]. Inkjet printing facilitates the controlled magnetic orientation of magnetic particles within, e.g., a polymer matrix- magnetic particle filler component by applying a magnetic field during printing. Such magnetic alignment results in reduction in hysteresis loss and increase in high frequency permeability [56,60]. In this technique, printing ink prepared by the dispersion of magnetic particles in a suitable fluid can be ejected by a inkjet print head onto a substrate [56]. To prepare the magnetic particles chemical or physical methods which provide particles in the nanometre range with a tight size distribution are employed. These particles should be coated prior to dispersion in the fluid to minimize agglomeration. A well dispersed magnetic fluid, also known as magnetic ink, is highly recommended since aggregation can result in the clogging of the inkjet device. Rheological parameters such as viscosity, surface tension, etc. of the

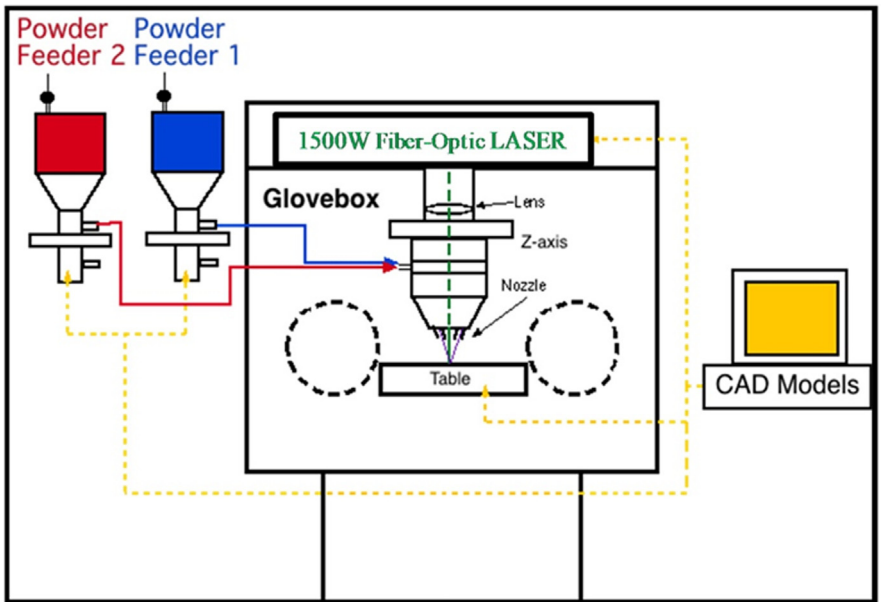


Fig. 3. Schematic of a Directed Energy Deposition setup.

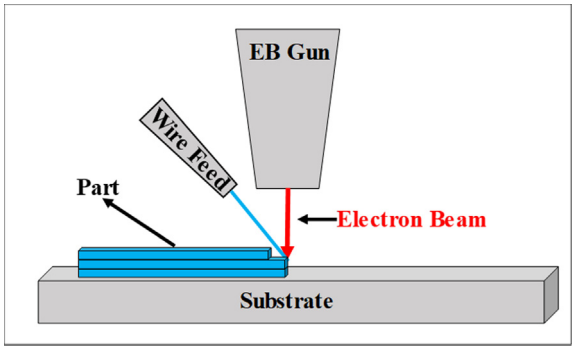


Fig. 4. Schematic of Electron Beam Additive Manufacturing system.

magnetic ink should be adjusted for better quality inkjet printing. Many complex patterns have been printed using inkjet printing [56,59]. Magnetic ink has been developed using metallic magnetic nanoparticles, such as cobalt [61,62], nickel [63,64], as well as oxide magnetic nanoparticles [65]. The applied magnetic field during inkjet printing results in alignment of the particles along their magnetic easy axis enhancing the magnetic anisotropy of printed component [56,66].

2.1.4. Sheet lamination

In this process, 2D metal sheets are stacked together to form a 3D object. A host of techniques have been used to join the sheets, e.g., brazing, laser welding, ultrasonic consolidation and diffusion bonding [67]. While there are some advantages of this process, e.g., low cost, good surface finish, ease of making large components, there are some significant limitations. Anisotropic properties are a major limitation of materials made by the sheet lamination process. Swelling of parts can also take place, leading to distortion along the z-axis.

2.2. Processing parameters and their effects on microstructure and magnetic properties

Processing parameters can have a significant impact on the microstructure and consequently the mechanical behaviour and magnetic behaviour of the alloys. Sames et al have provided a processing map for the case of metal AM and discussed the processing steps, parameters, and challenges [46].

2.2.1. Powder feedstock

The quality of the starting material plays a major role on the build properties. The shape, size, internal porosity, surface morphology, and composition are some of the relevant factors of the powder feedstock. The flowability of the powder, especially for DED

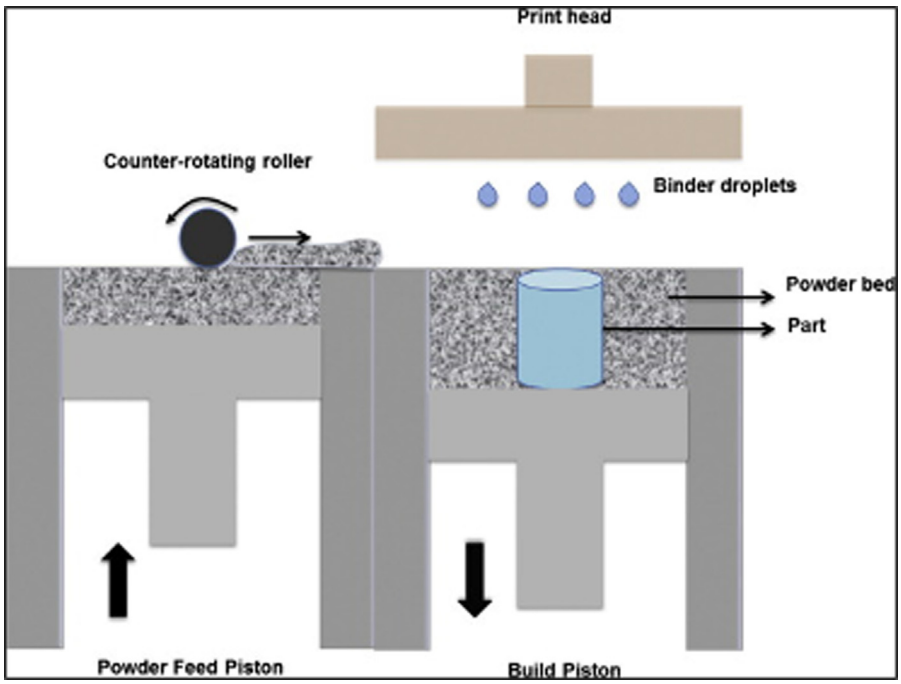


Fig. 5. Schematic of Binder Jet Process. Reprinted with permission from [55] © 2007, Elsevier.

techniques, depends on these factors. The quality of the powders largely depends on the technique used to make the powders. The most common techniques to make the powders are gas atomisation, rotary atomisation, plasma atomisation, and plasma rotating electrode process (PREP) [4,46,68]. PREP usually results in high quality powders [68,69]. The powder size used depends on the specific AM technique. PBF systems typically use powder particles in the size range of 20–50 μm, while most DED systems require larger sized particles (50–150 μm) for higher flowability. The smaller particles used for PBF systems tend to result in better surface finish as they do not have as many satellites as the larger powders used in the DED processes. However, there is usually a lower production yield in the manufacture of smaller particles, leading to increased cost [70]. Elemental blends have typically been used for the AM of magnetic materials [20,71]. The powders are mixed in a rolling mill for about 4–8 h prior to deposition to obtain good homogeneity. The use of elemental blends is more prominent in DED processes compared to PBF processes. While elemental blends have advantages to prepare alloys for which pre-alloyed powders are not readily available and for compositionally graded samples, compositional inhomogeneity in the build can be a challenge.

2.2.2. Effect of process parameters on defects, porosity, density, cracking, and magnetic properties

There are defects associated with the components produced by AM techniques. The most common of these defects is porosity. Porosity has a significant negative effect on the mechanical and magnetic properties. Porosity can arise from several factors, such as quality of powder particles, processing parameters and the nature of the solidification process. During the gas atomisation process, gases may be trapped in the powders, leading to porosity during deposition. Usually these pores are spherical and difficult to eliminate. The pores induced by the AM process are usually formed when the supplied energy is insufficient for complete melting of the powder particles. The input volume energy density (E_v) of the SLM process can be defined as [72]

$$Ev = \frac{P}{vht} \ \mathrm{J/m^3}$$

where P (W), v (m/s), h (m) and t (m) represent laser power, scanning velocity, hatch spacing and layer thickness, respectively. The energy density required for melting (E_m) a material can be estimated by the following equation [73]

$$E_m = c\rho(T_m - T_a)J/m^3$$

where c (J/kg K), ρ (kg/m^3), T_m (K) and T_a (K) represent specific heat, materials density, melting temperature and ambient temperature, respectively. The ratio of *Ev/Em* should be equal to or greater than 1 to melt the powder. Un-melted powder is one of the main reasons for a decrease in mechanical strength [74,75].

The density of the laser based AM processed components can be tuned by scan rate, laser power, hatch spacing, layer thickness and type of gas used in the chamber (Fig. 6) [76].

Defects can occur when process parameters are not optimized. Process induced porosity can be reduced by post processing techniques such as hot isostatic pressing (HIP). Cracking may arise during solidification or subsequent heating. The presence of pores can also lead to cracking. The process temperature plays a major role in defect formation. If the temperature is high, swelling of

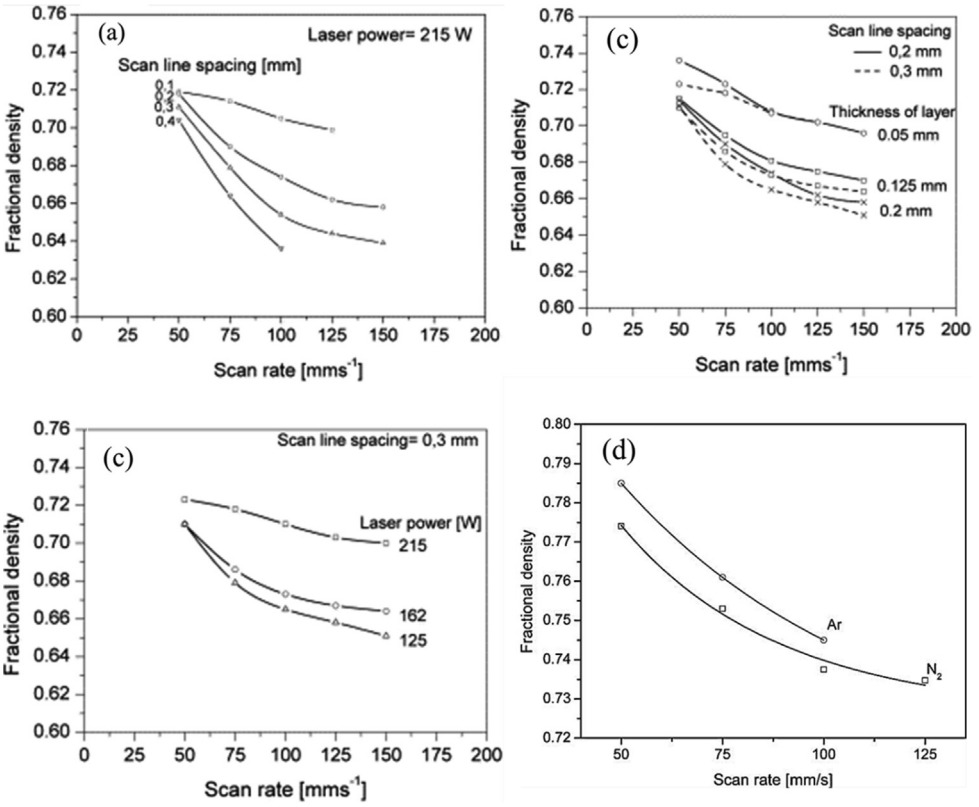


Fig. 6. Density of laser sintered iron as a function of scan rate for various values of (a) scan line spacing, (b) laser power, (c) thickness of layer and (d) chamber atmosphere. Reprinted with permission from [76] © 2003, Elsevier.

samples can occur, which causes interlayer cracking. This effect can be avoided by controlling the process parameters. The cracking is also highly material dependent [4,46]. Solidification cracking is usually caused when higher energy than required is applied, leading to higher thermal gradients, which can cause stresses and subsequent cracking. The other common mode of cracking is grain boundary cracking. Cracks nucleate along grain boundaries, especially in materials in which embrittling intermetallic precipitates form along grain boundaries. Liquation cracking is another mode of cracking which has been observed in AM processed samples.

The magnetic properties of AM processed samples depends on process parameters and the resulting microstructure due to the following reasons:

(i) A component prepared using a low energy density can have high porosity and defects associated with un-melted powders. High porosity results in low permeability, a lower magnetization per unit volume and higher demagnetizing field [77].
(ii) On the hand, a component prepared using a high energy density can have high density possibly together with cracking. Cracks can result in low mechanical strength.
(iii) The thermal gradient varies with time and position inside the melt pool, which can influence solidification rate, scale of microstructure, grain growth, etc. [78]. Usually, the larger the grains, the smaller the coercive force.
(iv) Cooling rate has a direct influence on the liquid–solid and solid–solid phase transformations, grain structure and hardness. The cooling rate is controlled by a combination of multiple process parameters, including the travel speed, the input power, thermal conductivity of the material and substrate, etc.
(v) Microstructural and related features such as grain size, texture, porosity, cracks, etc. influence magnetic domain wall pinning and thus the coercivity and permeability. Local laser melting and high cooling rate results in residual/internal stresses and minimal grain growth. Heating the building platform at optimum temperatures can result in lower residual/internal stresses. Hysteresis losses of AM processed sample can also be lowered by post processing heat treatments which have been shown to be effective in reducing the residual/internal stresses as well as promote grain growth.

3. Soft magnetic materials (SMM)

The materials which are easy to magnetise and demagnetise are classified as soft magnetic materials (SMM). Low coercivity (H_c), high saturation magnetization (M_s) and high permeability (μ) are some of the relevant parameters of such materials. Iron-silicon, iron-nickel, iron-cobalt, low carbon steel, iron, soft ferrites, etc. are some examples of SMM. The global market of SMM was ~$19

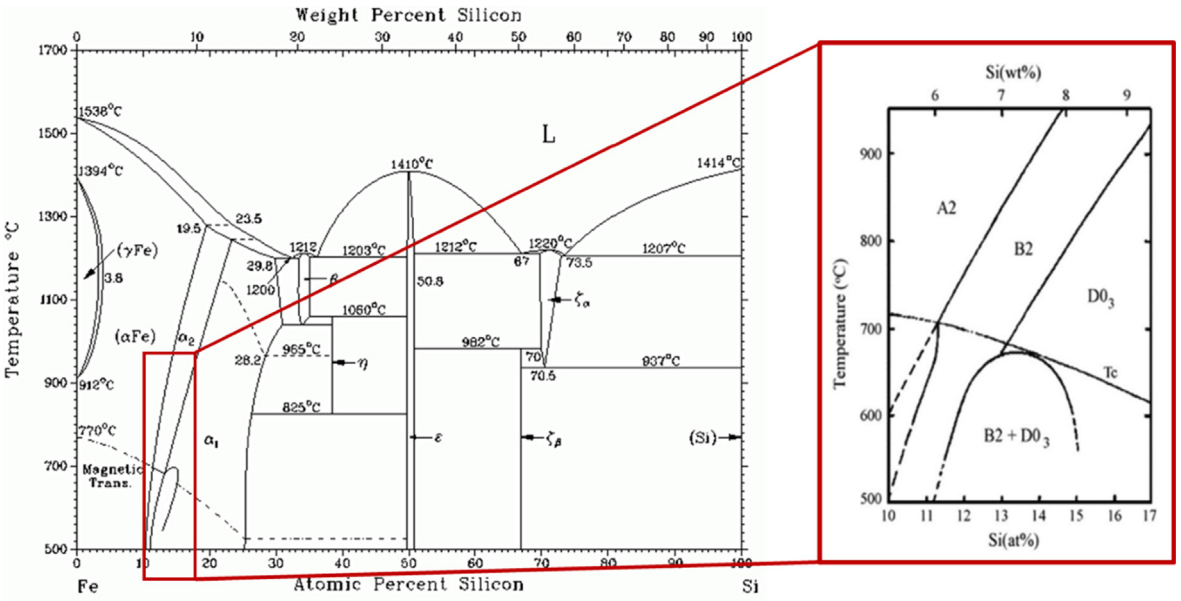


Fig. 7. Phase diagram of Fe-Si and the zoomed region of interest. Reprinted with permission from [84] © 1975, Taylor & Francis.

billion in 2016 and is expected to reach more than $42 billion by 2026 [79]. Metallic iron was one of the first SMM fabricated by laser-based AM techniques [76,80–82]. Since then, other SMM have been prepared by AM techniques [71,83]. Effects of AM processing parameters on the microstructure and their subsequent influence on the structural and functional properties have been investigated. SMM prepared by AM techniques are described in the following sections.

3.1. Fe-Si alloys

For electric motor applications, in addition to excellent soft magnetic properties, high electrical resistivity is also essential since eddy current losses are inversely proportional to electrical resistivity. High Si-steels have excellent electrical resistivity and good magnetic properties but are difficult to fabricate using conventional techniques because of their poor ductility. In conventional methods, the high porosity is also a major challenge when Si content is greater than ~4%. The variation in crystal structure and Curie temperature (T_c) with Si% content in the Fe-Si system is shown in Fig. 7 [84,85].

AM provides a process to produce high silicon containing steels. Fe-Ni-Si SMM were prepared by SLM using Ni coated silicon steel powder as the starting material [86]. With decreasing laser scan speed, the density increased from 50% to 99%, and the nature of the defects changed from porosity to cracking. The low laser scan speed is favourable for complete melting of the starting powder, resulting in higher M_s compared to samples prepared at higher scan speed. These SLM processed Fe-Ni-Si alloys showed anisotropic magnetic properties. A two-step melting process of Fe-Si alloys using SLM is helpful to increase the glassy phase fraction and reduce coercivity [87]. The effect of laser processing parameters on the magnetic properties of Fe-Si soft magnetic rings was also investigated [88,89]. The effect of laser energy and annealing temperature on micro-hardness is shown in Fig. 8.

The magnetic properties were found to be the best for rings built using a value of energy input that produced fibre texture, with the ⟨0 0 1⟩ axes aligned along the build direction [88]. This ⟨0 0 1⟩ fibre-texture can be changed to a cube-texture by increasing laser energy [91]. A recrystallized microstructure of mostly equiaxed grains was obtained after annealing at temperature of 1150 °C [89]. Fig. 9 shows the magnetic properties of ring samples before and after annealing, with a reference line indicating the magnetic properties for commercial grade of high-Si electrical steel. The maximum permeability (μ_{max}) of the SLM steel was found to be ~2000 and ~24,000 for as built and after annealing conditions, respectively. The trend of increasing resistivity (at constant slope) with temperature up to annealing temperatures of 900 °C may be associated with stress relief, as predicted for low-alloy steels. Annealing at high temperatures resulted in the lowest coercivity of 16 A/m [89]. The power loss decreased with increasing annealing temperature, from ~4.3 W/kg for the as-built sample to ~2.2 W/kg for samples annealed at 1150 °C. A similar trend was observed for the variation of H_c and power loss with annealing temperatures.

The high-silicon steel (6.9%wt.Si) prepared by SLM exhibited a ferrite phase without ordering, which is favourable for improved ductility [91]. Samples prepared by SLM exhibited very dense columnar grains oriented along the build direction. The microstructure of Fe-Si alloy parts prepared by SLM has been modified using a range of heat treatments [90]. The onset temperature for micro-segregation dissolution and grain growth was found to be ~700 °C and ~900 °C, respectively. After annealing for 1 h at 1150 °C, the grain size is greater than 100 μm; this grain size is close to the saturation grain size [90,92]. Annealing, followed by quenching could be promising to obtain a more homogeneous microstructure for magnetic applications [90].

The influence of laser energy input ($E = P/v$, P is the laser power and v is the scan speed) on the porosity and pore morphology are

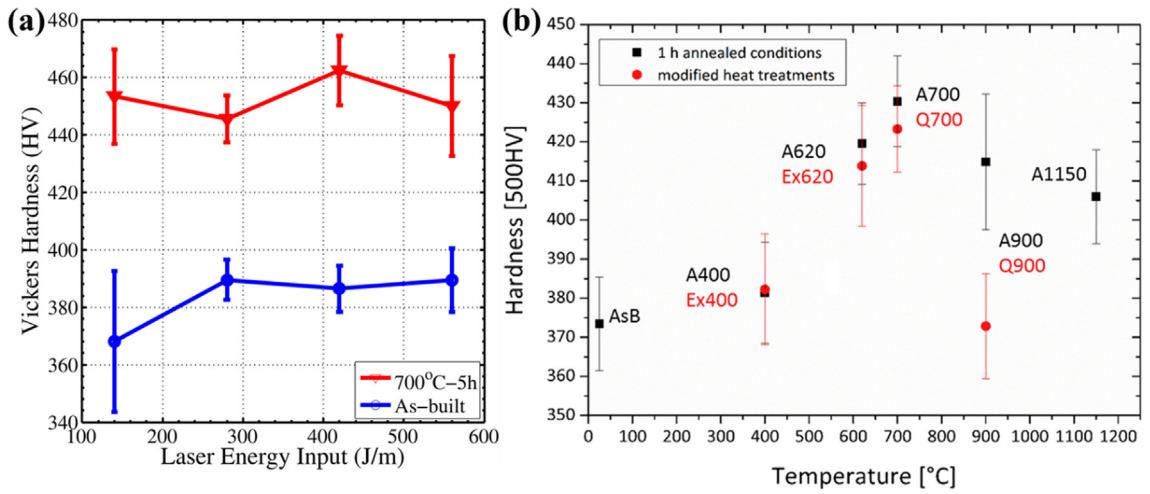


Fig. 8. (a) Hardness (HV) versus laser energy input for as built (blue line) and after heat treatment, heat treatment was performed to relieve the stress. Reprint with permission from [88] © 2018, Elsevier. (b) Hardness (HV) versus annealing temperature; black squares represent as built (AsB) and annealed for 1 h at 400 °C (A400), 620 °C (A620), 700 °C (A700), 900 °C (A900) and 1150 °C (A1150), while red dots represent either dwell time at 400 °C (Ex400), 620 °C (Ex620) or quenched from 700 °C (Q700), 900 °C (Q900). Reprinted with permission from [90] © 2017, Elsevier.

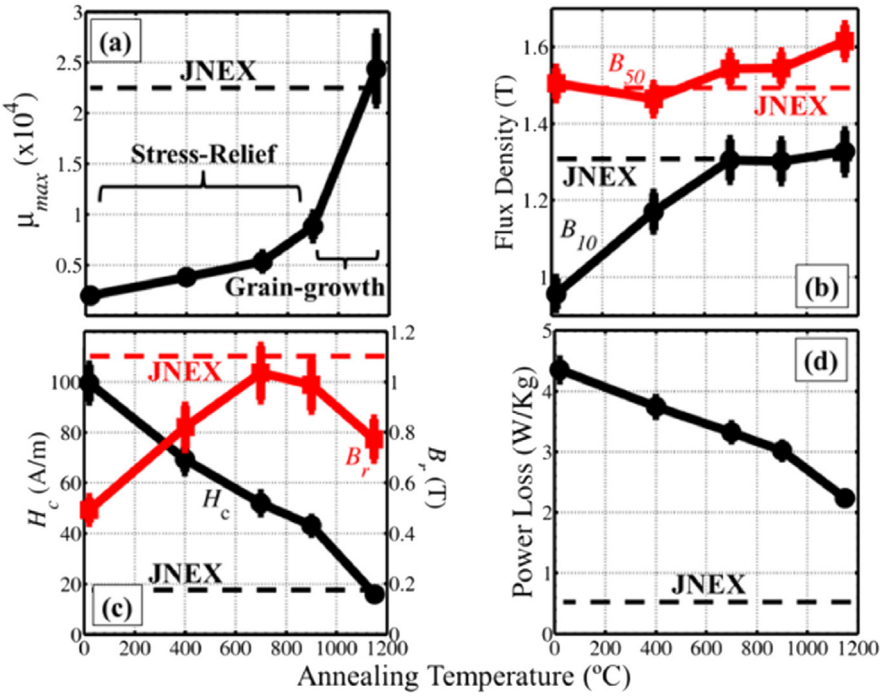


Fig. 9. Magnetic properties versus temperature for Fe-6.9%Si prepared by SLM (a) maximum magnetic permeability (b) flux density at magnetic field of 1000 A/m (B10) and 5000 A/m (B50) (c) coercivity and (d) power loss at B_{max} of 1 T and frequency of 50 Hz. The magnetic properties of a commercial grade of 0.1 mm thick high-Si electrical steel are shown by the associated dashed line. Reprinted with permission from [89] © 2018, Elsevier.

shown in Fig. 10 [91]. Samples with $E < 280$ J/m showed large pores of irregular shape. Increasing the value of E to 280 J/m resulted in a reduction in the number of irregular pores, however, crack formation was observed.

In another study, Fe-based bulk metallic specimens were printed by SLM from the gas-atomized $Fe_{68.3}C_{6.9}Si_{2.5}B_{6.7}P_{8.7}Cr_{2.3}Mo_{2.5}Al_{2.1}$ powder to study the effect of laser process parameters on the structure, thermal stability, and soft magnetic properties [93]. Low scan speed and high laser power resulted in the formation of high relative density (99.7%) samples (Fig. 11). The amorphous structure of the SLM processed samples was found to be similar to that of the starting powders.

$(Fe_{60}Co_{35}Ni_5)_{78}Si_6B_{12}Cu_1Mo_3$ soft magnetic alloys processed by the laser AM process exhibited comparable magnetic properties

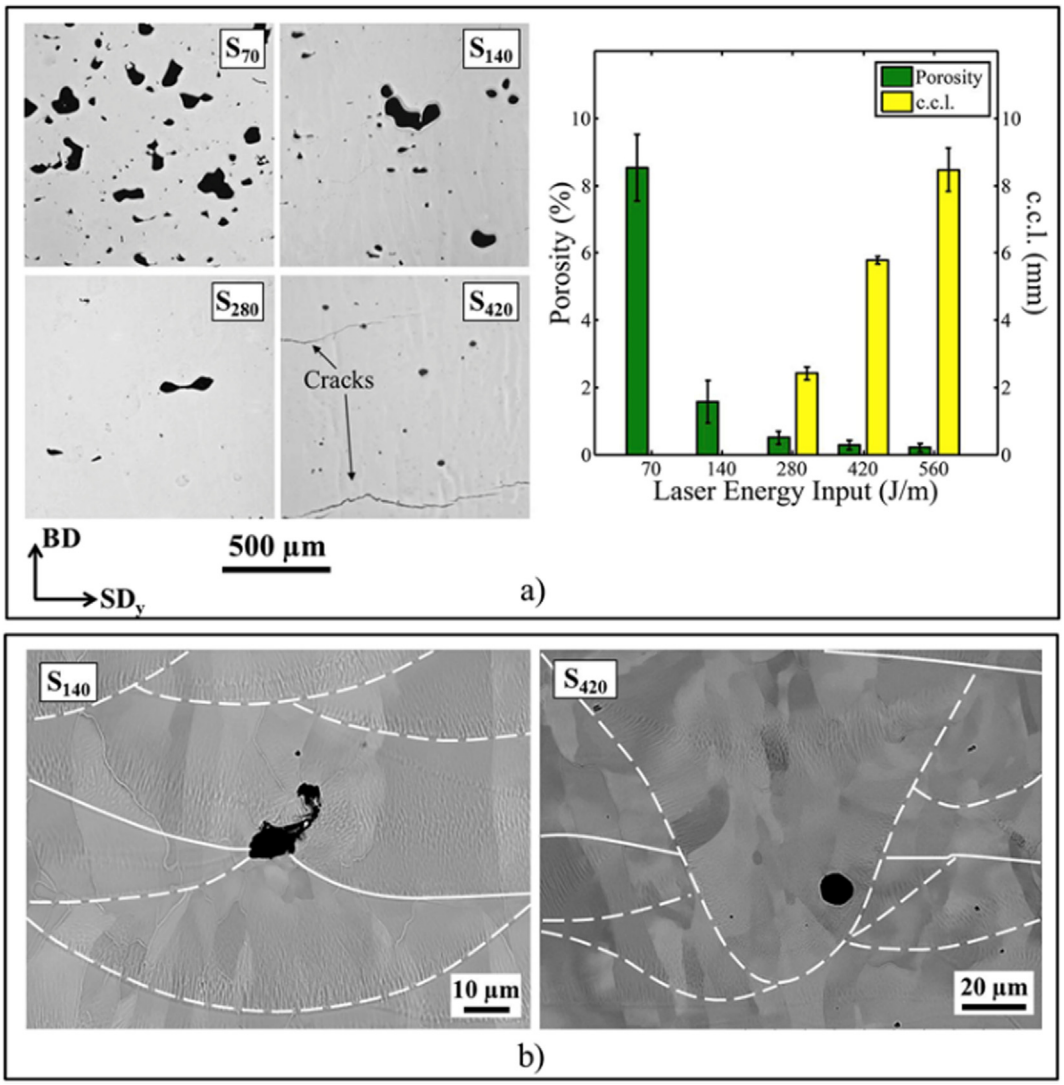


Fig. 10. Effect of laser energy input on porosity, crack formation and pore morphology. (a) optical micrographs, percentage porosity and cumulative crack length (c.c.l) versus laser energy input. (b) SEM of irregular (left) and spherical pores (right). Reprinted with permission from [91] © 2016, Elsevier.

to those prepared by conventional methods [94]. Microstructures with a higher volume fraction of the bcc phase exhibited substantially higher M_s. The volume fraction of the Fe–Si phase increased with increasing laser power, resulting in improved M_s. Alloys processed by a laser power of 3 kW exhibited M_s of 199 Am^2/kg and H_c of ~2387 A/m.

Recently, $(Fe_{60}Co_{35}Ni_5)_{73.5}Si_{13.5}B_9CuMo_3$ magnetic alloys, which contain higher Si content were prepared using laser AM and laser re-melting [95]. Laser re-melting was found to be useful to obtain improved magnetic and mechanical properties. The alloys possess a majority BCC crystal structure phase. A secondary phase (Fe_2B) was produced during solidification after re-melting. Laser re-melted alloys exhibited a M_s of 165.5 Am^2/kg and H_c of ~1671 A/m. A $FeSi_{6.7}$ alloy processed by the laser powder bed fusion technique showed very low magnetostriction, maximum permeability of 31,000, minimum coercivity of 16 A/m and hysteresis losses of 0.7 W/kg at 1 T and 50 Hz [96]. Thus, SLM is a promising technique to produce Fe-Si alloys with high Si%, with density and electrical resistivity higher than those of alloys prepared by conventional routes. Recently, Plotkowski et al reported the effect of scan pattern and sample geometry on the microstructure, magnetic properties and eddy current values of Fe-3Si components prepared by SLM [97]. A cross-section design based on the Hilbert space filling curve showed lower power losses in comparison to parallel plate construction and mesh structure.

3.2. Fe-Co alloys

Fe-Co alloys exhibit high saturation magnetization, high permeability, low magnetic losses and high Curie temperature, and are

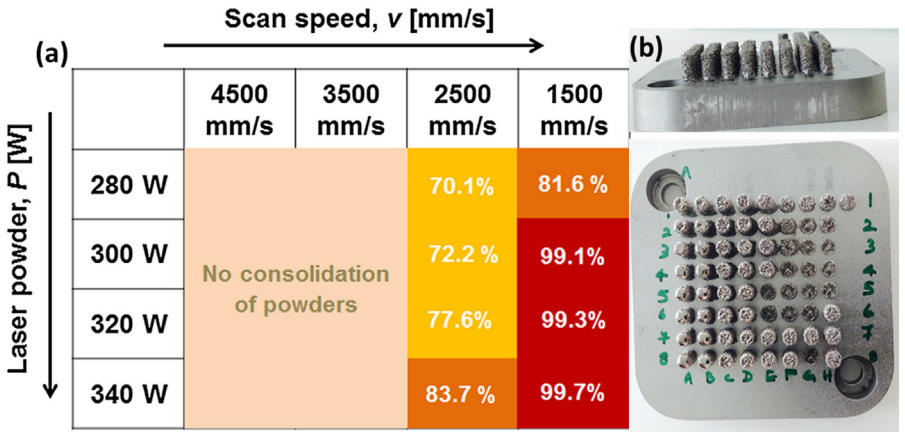


Fig. 11. (a) Change in relative density of samples processed by SLM with changing the laser power and scan speed, and (b) Photographs of SLM samples. Reprinted with permission from [93] © 2015, Elsevier.

therefore useful in applications where thermal stability is a primary factor. Alloying of Fe-Co with V [98,99], Cr [100,101], Ni [100,102], Mn [103], W [104], Nb [98,105,106], Si [107], Al [108] has been studied to determine their effect on structural, mechanical and/or magnetic properties. The development of Fe-Co soft magnetic alloys has been documented in reviews published e.g., in *Progress in Materials Science* [109,110]. The phase diagram of Fe-Co is useful to determine the annealing/sintering temperatures needed to obtain the phase(s) of interest (Fig. 12) [111]. On the Fe rich side, the Fe-Co alloys exhibit bcc based (α and α_2) phases up to ~900 °C and the fcc (γ) phase at higher temperatures. On the Co rich side, the fcc crystal structure is stable at temperatures up to the melting temperature.

We now review AM processed Fe-Co alloys. Fe-Co alloys produced by SLM showed mechanical properties similar or better to a commercial reference material [112,113]. The compositional dependence of the magnetic properties of a range of Fe-Co alloys compositions prepared by the LENS AM technique was in good agreement with those of samples prepared using conventional methods [20,38].

Alloying with V improves ductility while retaining good magnetic properties. A bulk Fe-Co-1.5 V alloy prepared by LENS showed good ductility with no significant macroscopic defects (Fig. 13) [114]. The induction of as built and annealed samples was found to be 2.23 T and 2.29 T, respectively. The value of H_c decreased from 995 A/m for as built samples to 401 A/m for annealed samples. This change was probably due to finer grain size and/or retained residual stress/strain in the as-built condition. AM processing parameters, such as build velocity and interlayer interval time, can be used to control the relative fraction of the B2 phase in Fe-Co-1.5 V alloys [115]. The hardness of LENS processed Fe-Co-1.5 V alloys was found to be significantly higher than conventionally processed bulk material.

The magnetic properties of LENS deposited samples were not as good as those produced by conventional methods. However,

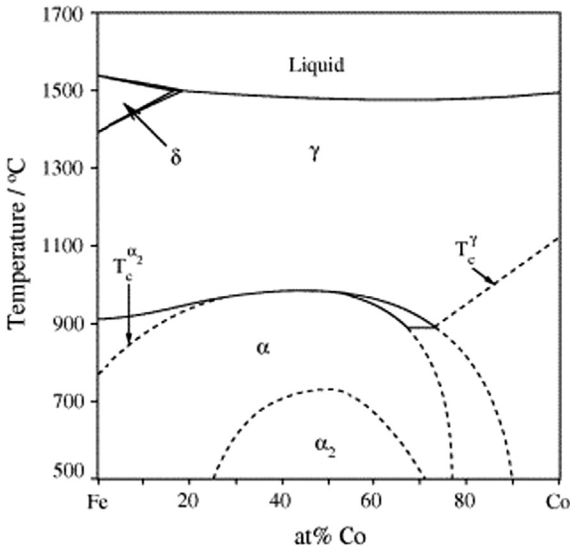


Fig. 12. The Fe–Co binary diagram [111].

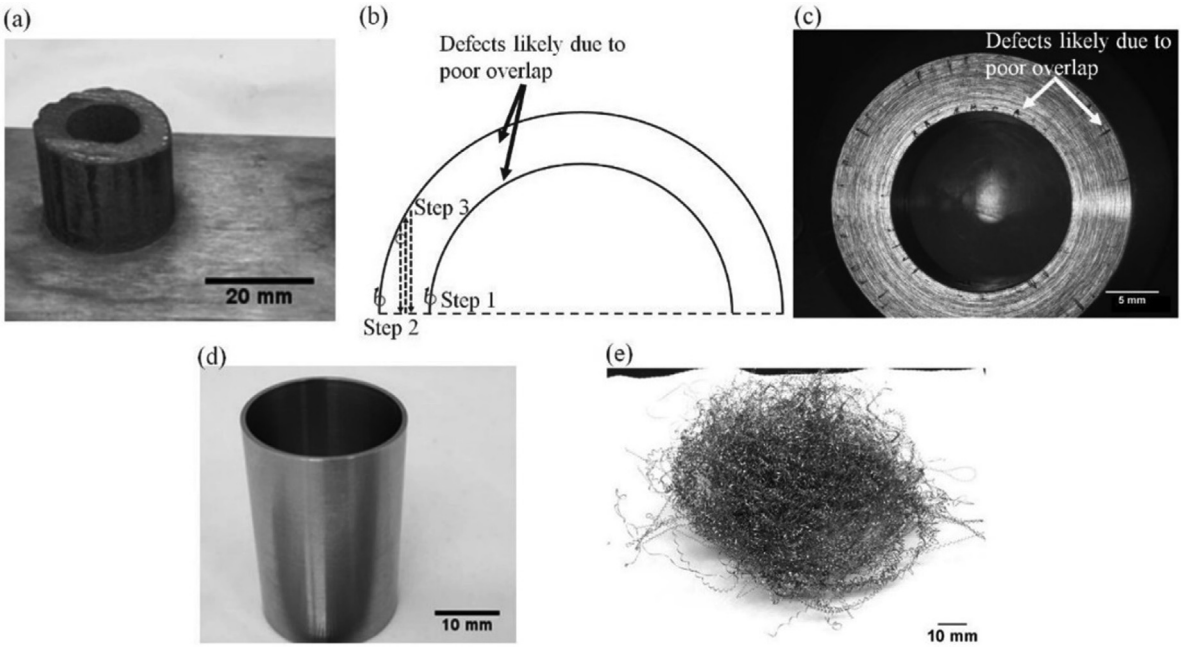


Fig. 13. (a) A cylinder of Fe-Co-1.5 V cylinder built by LENS AM, (b) Crosshatch build pattern, (c) Defects formation along the edge, (d) Thin-wall cylinder geometry produced by layer-by-layer deposition of concentric circles of material exhibited relatively less defects, (e) Long continuous chips produced from thin wall cylinder geometry exhibited better ductility. Reprinted with permission from [114] © 2018, Elsevier.

optimization of processing and post processing conditions could potentially improve these properties.

Riipinen et al. studied the magnetic, electrical and mechanical behaviour of AM processed Fe-49Co-2 V alloys [116]. Various heat treatments were used to alter the magnetic properties. As can be seen from the Table 1, the magnetic properties improved with heat-treatment. Samples heat treated at 750 °C/2h + 820 °C/10 h exhibited large grain size, low coercivity and high permeability. A homogenous microstructure with large grain size was found to be essential for low magnetic coercivity and high permeability [109].

3.3. Fe-Ni alloys

Fe-Ni alloys are well-known soft-magnetic materials due to their high permeability, low coercivity, low magnetostriction and substantial anisotropic magnetoresistance [117,118]. Fe-Ni alloys with ~20% Fe and ~80% Ni have been used for decades as soft magnetic materials in numerous commercial products [117]. Fig. 14 shows the magnetic properties of permalloy and other soft magnetic materials [119]. Typical applications of Fe-Ni alloys are in transformer laminations, magnetic recording head sensors, magnetic shielding, magnetic sensors, magnetic amplifiers, power transformers, etc. [120,121]. Substantial research to use Fe-Ni based alloys in the field of magnetic cooling has also been performed since they exhibit higher relative cooling power and low coercivity compared to first order, rare earth based, expensive magnetocaloric materials [122–128]. Alloying with small amounts of Cu, Mo, Cr can improve ductility with minimal change in magnetic properties [121,124].

AM has been used to process Fe-Ni based alloys [20,36,71,129–139]. Mikler et al. reported the fabrication of Fe-30Ni, Ni-15Fe-5 V

Table 1
Magnetic properties of SLM processed and conventionally processed alloys prepared from gas atomized Fe-49Co-2 V powder and commercial Fe-49Co-2 V alloy [116]. H_c, μ_{max} and B_{max} stand for coercivity, permeability and saturation magnetization, respectively.

3D Printed	Shape	Heat-treatment	H_c (A/m)	μ_{max}	B_{max} (T)
	Bar	As-built	1345	307	2.17
	Ring		1288	322	1.57
	Bar	820 °C/4h	73	9156	2.29
	Ring		493	1922	2.18
	Bar	750 °C/2h + 820 °C/10 h	52	17,000	2.28
	Ring		47	13,000	2.23
	Bar	950 °C/0.5 h + 820 °C/10 h	527	1421	2.30
	Ring		144	4463	2.16
Commercial	Bar	As-built	85	2204	2.23
	Ring		104	1811	2.18
	Bar	820 °C/4h	59	4300	2.24
	Ring		40	12,000	2.23

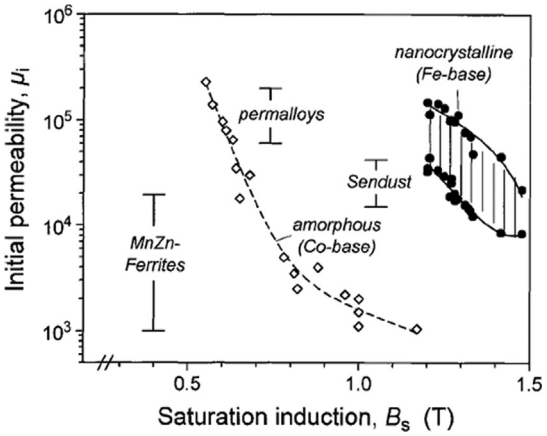


Fig. 14. Initial permeability and saturation induction of well soft magnetic materials. Reprinted with permission from [119] © 1997, Elsevier.

and Ni-15Fe-5Mo alloys using the LENS process. The change in laser processing parameters can significantly influence the microstructure and magnetic properties [71,132,133]. The microstructure of a Fe-30Ni alloy was tuned from a majority bcc phase to a majority fcc phase by changing laser processing parameters. Higher travel speeds resulted in greater fcc phase content. The M_s of bcc phase dominated samples was found to be higher than those of fcc phase dominated samples. The structural and magnetic properties of Fe-Ni based soft magnetic materials prepared by AM are shown in Table 2. At lower travel speed of 8 in./minute (IPM), M_s values decreased and H_c increased with decreasing laser power. For a travel speed of 13 IPM, M_s values increased, and H_c decreased with decreasing laser power [36] (Fig. 15). The M_s variation in Fe-30Ni is due to a difference in total magnetic moment for the fcc phase (1.15 μB/atom) and the bcc phase (1.87 μB/atom) [140].

The M_s values of AM processed and conventionally processed Fe-Ni based alloys were found to be comparable. However, H_c of AM processed Fe-Ni based alloy was higher than that of commercial permalloy. Usually, a low value of H_c is the result of very carefully selected heat treatment schedules, on the other hand, only as-built AM processed Fe-Ni alloys have been studied in detail. Appropriate heat treatment of the as-deposited alloys may result in a significant reduction in H_c values. The Fe–30Ni alloy was also

Table 2
Saturation magnetization (M_s) and coercivity (H_c) values of Fe-Ni based soft magnetic materials as a function of laser processing parameters. IPM represents inches per minute.

Processing condition	Phase(s)	M_s (Am^2/kg)	H_c (A/m)	Refs.
Fe-80Ni				
P = 100 W, v = 0.1 m/s	$FeNi_3$	99	~2865	[130]
P = 100 W, v = 0.2 m/s	$FeNi_3$	~98	~2785	
P = 100 W, v = 0.3 m/s	$FeNi_3$	~96	~2865	
P = 100 W, v = 0.4 m/s	$Fe_{70}Ni_{30}$ + Ni	95	~2944	
Fe–30Ni				
P = 300 W, v = 5 IPM	Majority of FCC	120	~2387	[133]
P = 300 W, v = 20 IPM	Majority of BCC	165	~2059	
P = 500 W, v = 5 IPM	FCC	125	~1512	
P = 500 W, v = 20 IPM	Majority of BCC	152	~1671	
P = 300 W, v = 8 IPM	Majority of FCC	95	~3740	[71]
P = 400 W, v = 8 IPM	Majority of FCC	99	~1034	
P = 500 W, v = 8 IPM	FCC	105	~796	
P = 300 W, v = 13 IPM	Majority of FCC	114	~1194	
P = 400 W, v = 13 IPM	Majority of FCC	103	~2546	
P = 300 W, v = 13 IPM	FCC + BCC	96	~3501	
P = 110 W, v = 0.1 m/s	Fe_7Ni_3 and Fe_3Ni_2		~350	[131]
P = 110 W, v = 0.4 m/s	Fe_7Ni_3 and Fe_3Ni_2		~80	
P = 110 W, v = 1.6 m/s	Fe_7Ni_3 and Fe_3Ni_2		~342	
Ni-15Fe-5 V				
P = 400 W, v = 20 IPM	FCC	72	~477	[132]
Ni-15Fe-5Mo				
P = 400 W, v = 20 IPM	FCC	82	~390	[132]
P = 200 W, v = 0.4 m/s + annealing	FCC	78	~120	[141]
Ni-21.5Fe				
E = 428.57 J/mm^3	$FeNi_3$	97		[135]
E = 71.43 J/mm^3	$FeNi_3$	94.8		

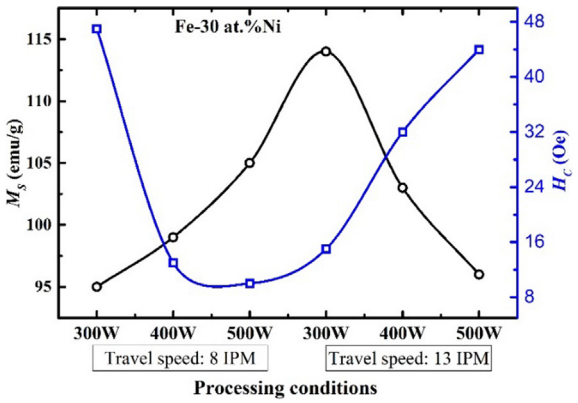


Fig. 15. Saturation magnetization (M_s) (left axis) and coercivity (H_c) (right axis) as a function of processing conditions (travel speed and power) for Fe-30Ni alloys.

prepared using the SLM process [129]. Lower bcc phase lattice parameter and large grain size was obtained for high laser scan velocity and low power. H_c was found to be 88 A/m at low laser scan velocity of 400 mm/s and laser input power of 110 W.

Schönrath et al. studied the magnetic properties of a $Ni_{78.5}Fe_{21.5}$ alloy processed by AM [135]. The M_s of AM processed alloy was found to be higher than that of the initial powder blend.

3.4. Soft magnetic composites

Soft magnetic composites (SMCs) are of high interest due to their useful properties such as low core loss, high saturation magnetization, and high electrical resistivity [142–144]. They have applications in energy generation and conversion, and sensing. Various SMCs such as Fe–Si/MnZn ferrite [145], Fe-Si-Al/MnZn ferrite [146], Fe-Si-Al/NiZn ferrite [147], Fe-Si/ZrO_2 [148] Fe-Cu-Nb-Si-B [149], etc. have been studied. Usually, SMCs have been developed using powder metallurgy methods [149,150].

Fe-Si based SMCs have been prepared using thermal spray [151–153], plasma spray [154] and cold spray [155–158] technologies. SMCs fabricated by the cold spray technique using Ni-coated FeSiAl composite powder exhibited H_c values of ~4774 A/m [155]. Iron powder coated with phosphorus has been printed using binder-jet printing and direct metal laser sintering [159]. The samples prepared by binder-jet were cured at 815 °C to burn off the binder and then annealed at 1260 °C for densification.

Carbonyl iron particles filled thermoplastic composites were fabricated by fused deposition modelling (FDM) [160]. These composites exhibit low remanence (M_r) and coercivity. The M_s and M_r of the composites increased with increasing carbonyl iron powder content. An acrylonitrile butadiene styrene (ABS) polymer matrix with stainless steel micro powder as the filler was used to develop a polymer-soft magnet composite [161]. Cube shaped samples of an iron-polymer composite filament were prepared using FDM. Hysteresis loops were obtained for magnetic fields applied along the [1 0 0], [1 1 0], and [0 0 1] directions of the cubes [162]. All the samples showed an easy plane along the layer orientation, and a hard axis vertical to the printed layers irrespective of the process parameters used for printing.

A magnetic field assisted projection stereolithography (MPSL) AM has been developed for 3D smart polymer composites [163,164]. In this process, nano or micro sized magnetic particles were deposited in the liquid polymer using a programmable micro-deposition nozzle. An external magnetic field can be used to control the magnetic particles at the required position and orientation. A digital mask image can then be employed to cure the photopolymer.

3.5. Soft magnetic ferrites

Iron oxide is a ferromagnetic ceramic exhibiting higher electrical resistivity and lower saturation magnetization than typical metallic magnetic materials. High resistivity of these ceramics makes them very useful for high frequency applications. Recently, UV-curable NiZn- and NiCuZn-ferrite pastes were developed for 3D printing [165,166]. Increasing the UV-sensitive monomer content resulted in faster curing, however, relative permeability was lower. A core structure ~4 mm tall was prepared by a 3D printer in a short time using an UV-curable feedstock. Ferrite cores with relative permeability up to ~103 and resonance frequency >30 MHz were achieved by altering the composition of the magnetic paste and sintering temperature [165]. After sintering, the inductance and resistance of the 3D-printed planar inductor were 792 nH and 15 mΩ, respectively [166]. These values of inductance and resistance are close to the values predicted by 3D finite element analysis.

4. Graded magnetic alloys

Graded soft magnetic alloys can be useful to rapidly discover new compositions with better properties, *via* a combinatorial approach, as well as to design engineering parts with functionally graded and locally optimized properties. Systematic variation in composition [20,34,36,167], grain size [168], precipitate density [169], twinning [170], etc. can be used to tune the properties.

Table 3
Saturation magnetization (M_s), coercivity (H_c) and hardness values of Fe-Si, Fe-Co and Fe-Ni based soft magnetic materials prepared by AM.

Laser process parameters/composition	M_s (Am^2/kg)	H_c(A/m)	Hardness (Hv)	refs
Fe-Ni-Si alloys (SLM)				
v = 1.2 m/s	~91.8	~2148		[86]
v = 2 m/s	~91.3	~2228		
v = 4 m/s	~86.4	~2148		
Fe–Si–B–Nb–Cu graded alloy (LENS)				
$Fe_{74.4}Si_{3.8}B_{17}Nb_{3.5}Cu_{1.3}$	150	~3820		[37]
$Fe_{68.9}Si_{18}B_{10}Nb_2Cu_{1.1}$	143	~1112		
$Fe_{64.3}Si_{24}B_8Nb_{2.7}Cu_1$	118	~1273		
$Fe_{73.5}Si_{13.5}B_9Nb_3Cu_1$ (LENS)				
v = 20 IPM, P = 340 W	143	~1671		[71]
v = 20 IPM, P = 430 W	151	~2069		
v = 40 IPM, P = 340 W	140	~3023		
v = 40 IPM, P = 430 W	146	~1989		
v = 60 IPM, P = 270 W	139	~1194		
$(Fe_{60}Co_{35}Ni_5)_{78}$ $Si_6B_{12}Cu_1Mo_3$ (SLM)				
v = 5 mm/s, P = 2000 W	179	~1671		[94]
v = 5 mm/s, P = 2500 W	191	~2467		
v = 5 mm/s, P = 3000 W	199	~2626		
$(Fe_{60}Co_{35}Ni_5)_{73.5}Si_{13.5}B_9CuMo_3$ (SLM)				
v = 20 mm/s, P = 1400 W	151.7	~4616	~875	[95]
v = 20 mm/s, P = 1600 W	162.9	~3978	~875	
v = 20 mm/s, P = 1400 W/2000 W	152.8	~1114	975	
v = 20 mm/s, P = 1600 W/2000 W	165.5	~995	975	
$Fe_{71}Si_{10}B_{11}C_6Cr_2$ (SLM)				
Powder used for printing	~140	99		[87]
P = 120 W	~140	1032		
P = 120/120 W	~140	397		
P = 120/120 W/annealing	~140	238		
$Fe_{68.3}C_{6.9}Si_{2.5}B_{6.7}P_{8.7}Cr_{2.3}Mo_{2.5}Al_{2.1}$ (SLM)				
Powder used for printing	104.2	~53		[93]
v = 2500 mm/s, P = 340 W	103.6	~33		
v = 1500 mm/s, P = 340 W	102.8	~28		
v = 170 mm/s, P = 300 W	**Fe-Co (LENS)**			[20]
$Fe_{30}Co_{70}$	199.3	~3199	220	
$Fe_{40}Co_{60}$	201.7	~4416	240	
$Fe_{55}Co_{45}$	216.7	~4432	260	
$Fe_{60}Co_{40}$	223.7	~3247	240	
$Fe_{70}Co_{30}$	248.3	~3112	205	
v = 170 mm/s, P = 300 W	**Fe-Ni (LENS)**			[20]
$Fe_{30}Ni_{70}$	119.8	~438	150	
$Fe_{40}Ni_{60}$	129	~358	160	
$Fe_{55}Ni_{45}$	161.6	~390	140	
$Fe_{60}Ni_{40}$	165.6	~493	145	
$Fe_{70}Ni_{30}$	168.7	~629	122	

Some types of electric motors require soft magnets, conductors, insulators and hard magnets. Fixing hard magnets at appropriate positions with an adhesive on a rotor can be used in surface mounted permanent magnet motors. Alternatively, permanent magnets can be fixed inside the rotor (interior permanent magnet motor). These techniques have several drawbacks, e.g., poor mechanical properties, especially at high speeds or elevated temperatures. Using AM, it is possible to produce compositionally/functionally graded rotors with an appropriate variation of properties with location.

4.1. Graded Fe-Si alloys

A high throughput approach for evaluating the microstructure and magnetic properties of compositionally graded Fe–Si–B–Nb–Cu alloys processed by the LENS technique was reported [37]. The microstructure of such alloys with lower Si/B content exhibited dendritic α-Fe_3Si grains, with B and Nb partitioning to the inter-dendritic regions. As the Si/B content increases, a (Fe, Nb) enriched eutectic phase was observed along with α-Fe_3Si grains, no Fe_3B was observed. These microstructural changes with varying Si/B ratio influenced the magnetic properties of these laser-deposited alloys. Table 3 shows the change in magnetic and mechanical properties of AM processed soft magnetic materials. (See Table 4.).

17-4PH stainless steel processed by conventional routes is mainly martensitic at room temperature, with less than 10 wt%

Table 4
Structure and hardness of $Al_xCoCrFeNi$ HEAs prepared by AM and conventional methods.

Alloy	Crystal structure	Hardness	Processing method	Ref.
Alloy library of $Al_xCoCrFeNi$				[206]
x = 0.15–0.37	FCC	~4 to 5 GPa	Optomec MR-7 LENS™, P = 150 W R = 12.7 mm/s	
x = 0.41–0.48	FCC + BCC	~5 GPa (FCC)		
		~8 GPa (BCC)		
x = 0.45–1.06	FCC + BCC/B2	~8 GPa (BCC)		
x = 1.16–1.32	BCC/B2	~8 GPa (BCC)		
AlCoCrFeNi	BCC/B2	~543 (HV0.5)	Optomec MR-7 LENS™, R = 2.5–40 mm/s	[229]
AlCoCrFeNi	BCC/B2	–	SLM: Concept Laser Mlab	[223]
AlCoCrFeNi	BCC/B2	~570 Hv to ~630 Hv	SLM: Farsoon FS271 M	[230]
$Al_xCoCrFeNi$			TRUMPF TruLaser Cell 7040 coaxial blown powder laser deposition	[208]
x = 0.3	FCC	~2.5 GPa		
x = 0.6	FCC + BCC	~3 GPa		
x = 85	BCC	~6 GPa		
$Al_xCoCrFeNi$			Arc Melting	[228]
x = 0.1–0.4	FCC	120 Hv	As cast	
x = 0.5–0.8	FCC + BCC	~200 to ~400 Hv		
x = 0.9–2	BCC	527 to ~510Hv		
$Al_xCoCrFeNi$			Arc Melting	[231]
x = 0.1	FCC	1.83 GPa	As cast	
x = 0.75	FCC + BCC	4.57 GPa		
x = 1.5	BCC/B2	5.60 GPa		
$Al_xCoCrFeNi$			Arc Melting	[224]
$0 \le x < 0.45$	FCC		As cast, double phase reason extended after homogenization	
$0.45 \le x \le 0.88$	FCC + BCC			
$0.88 < x \le 2$	BCC			
AlCoCrFeNi	BCC/B2		Induction melting	[223]

austenite [171]. However, 17-4PH prepared by SLM exhibited a higher proportion (~80 wt%) of austenite [172–174]. The coupled effect of rapid solidification condition and subsequent solid-state re-heating of the deposited layers in SLM-built 17-4PH presumably aids in the formation of the austenite phase. Recently, Freeman et al. used SLM to produce microstructurally and magnetically graded 17-4PH samples by manipulating the solid-state austenite–martensite phase transformation [175]. Fully austenitic regions exhibited paramagnetic properties, while dual-phase regions showed an increase in ferromagnetic behavior with increasing martensite mass fraction (Fig. 16).

A magnetic core made by materials with a tunable permeability can result in a more uniform flux distribution [176]. A three permeability toroid (TPT) core, with increasing permeability from the inner to the outer ring, was manufactured to obtain flux uniformity. The volume of TPT core was smaller than that of a single permeability toroid core. A 3D-printed TPT core had higher inductance than the corresponding value for a single-permeability core.

4.2. Graded Fe-Co and Fe-Ni alloys

Recently, we reported the structural, magnetic and mechanical properties of functionally graded Fe_xCo_{100-x} (x = 30–70) and Fe_xNi_{100-x} (x = 30–70) soft magnetic materials processed by laser AM (LENS) from a feedstock of elemental powders [20]. Fe_xCo_{100-x} showed a bcc structure in the composition range studied, while $Ni_{100-x}Fe_x$ exhibited either single phase fcc or a mixture of fcc and bcc phases, depending on the composition. The M_s for both Fe_xCo_{100-x} and Fe_xNi_{100-x} compositionally graded alloys increased monotonically with increasing Fe content while the H_c variation is not monotonic. The H_c values of all the compositions in $Ni_{100-x}Fe_x$ were

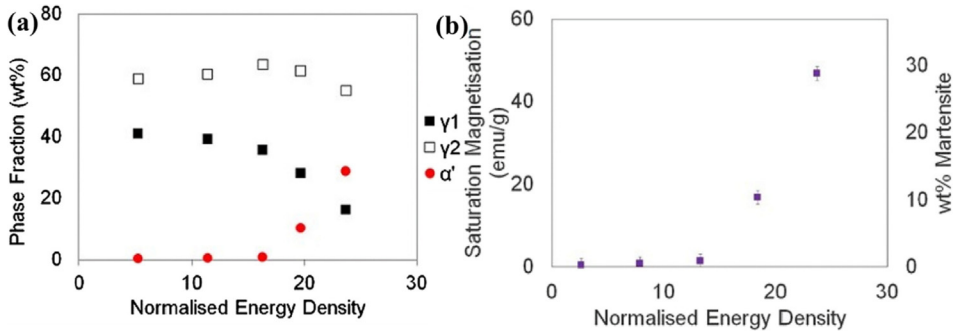


Fig. 16. (a) Variation in relative phase fractions of two austenite phases (y1 and y2) and martensite phase (α^1) with energy density. (b) dependence of saturation magnetization on phase fraction of martensite phase and energy density. Reprinted with permission from [175] © 2019, Elsevier.

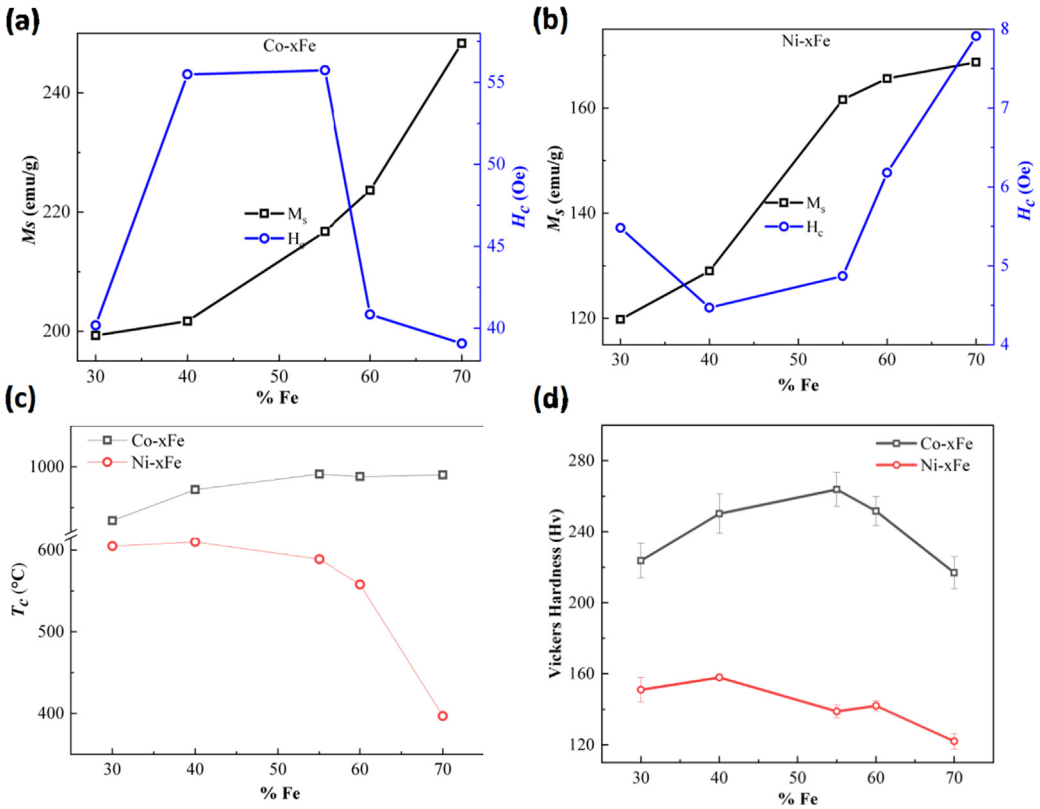


Fig. 17. Saturation magnetization (M_s) (left axis, black square) and coercivity (H_c) (right axis, blue circle) as a function of Fe concentration for (c) Co-xFe and (d) Ni-xFe compositionally graded alloys, Change in (c) Curie Temperature (T_c) and in (d) hardness as a function of Fe concentration for Co-xFe (black square) and Ni-xFe (red circle) compositionally graded alloys. Reprinted with permission from [20] © 2020, Elsevier.

found to be less than 630 A/m. In Fe_xCo_{100-x}, $Fe_{60}Co_{40}$ with M_s = 223.7 Am^2/kg, H_c = ~3247 A/m, T_c = 988 °C and hardness = 240VHN, and in Fe_xNi_{100-x}, $Fe_{55}Ni_{45}$ with M_s > 160 Am^2/kg, H_c < 400 A/m, Tc = 580 °C and hardness of 140 VHN, are the compositions which exhibited the best combination of magnetic and mechanical properties. Magnetic and mechanical properties of compositionally graded Fe_xCo_{100-x} and Fe_xNi_{100-x} alloys are shown in Fig. 17. A rapid approach was used to measure the T_c by employing a permanent magnet placed near a Thermogravimetric analysis (TGA) furnace. The magnet pulls the sample with a force proportional to the magnetic permeability of the sample. At the temperature where the sample property changes from ferromagnetic to paramagnetic (T_c), an apparent weight change will be observed. This method to measure the T_c is faster and cost effective than the conventional method using a VSM, PPMS, or MPMS, and therefore, recommended especially for rapid assessment of big alloy library.

4.3. Graded high entropy alloys

High entropy alloys (HEA) have attracted high interest due to their unique properties [177–179]. These properties depend on the alloy composition and structure [34,180–197]. There are numerous papers on the structural and mechanical properties of HEAs prepared by AM [190,191,198–221] but much less effort has been devoted to functional properties [34,35]. $AlCo_{1-x}Cr_xFeNi$ and $Al_xCrCuFeNi$ HEAs have been developed which exhibit a variation in magnetic properties for a range of x values. Some compositions exhibited H_c higher than 8000 A/m, which can be characterized as semi hard while others had H_c values less than 2400 A/m. The Curie temperature can also be tuned by changing the composition of these HEAs.

Graded $Al_xCrCuFeNi_2$ ($0 \le x \le 1.5$) HEAs were deposited using the LENS process employing a blend of elemental powders [35]. There was a gradual change in the structure from fcc to fcc + L12 to fcc/L12 + bcc/B2 to bcc/B2 with increasing Al content in $Al_xCrCuFeNi_2$. This change was accompanied by an increase in microhardness (Fig. 18). The higher Al content side of compositionally graded $Al_xCrCuFeNi_2$ possessed the bcc/B2 phase and was substantially harder than the fcc/L12 structure obtained for lower Al content Fig. 17.

In $Al_xCuCrFeNi_2$ HEA, the M_s was initially low for lower values of x, however, it increased rapidly to a maximum value when x = 1.3, and then decreased beyond x = 1.3 (Fig. 19) [35]. This change in magnetic properties of HEA can be explained based on the structure and phase stability. Lower Al content compositions ($0 < x < 0.8$) contained a negligible fraction of the B2 (ferromagnetic) phase and therefore exhibited a low value of M_s. Al-rich compositions which exhibited a significant fraction of B2 phase showed substantially higher M_s values.

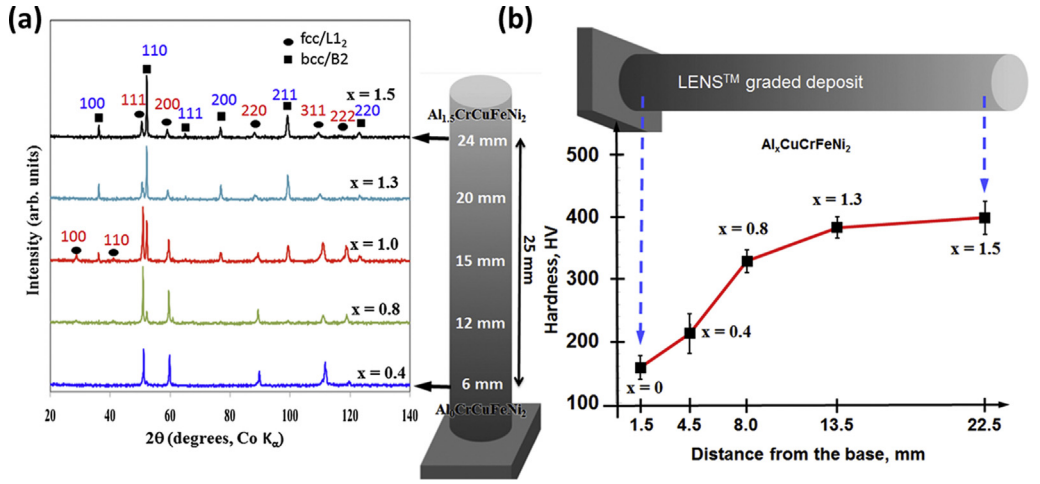


Fig. 18. (a) XRD data and (b) Vickers micro-hardness of compositionally graded $Al_xCuCrFeNi_2$ HEA. Reprinted with permission from [35] © 2017 Elsevier.

The microstructure and magnetic properties of compositionally graded $AlCo_xCr_{1-x}FeNi$ ($0 \leq x \leq 1$) HEA fabricated via LENS [34] were also studied. Fig. 20 shows the change in composition and microstructure along the cylinder height. XRD results revealed a mixture of B2 and bcc phases [34]. There was not much change in the phases present in $AlCo_xCr_{1-x}FeNi$ ($0 \leq x \leq 1$) HEA by changing the Co/Cr ratio. Transmission electron microscopy and atom probe tomography analysis of the $x = 8$ alloy showed spinodal decomposition within the BCC grains of this alloy.

The M_s of $AlCo_xCr_{1-x}FeNi$ ($0 \leq x \leq 1$) increased six times in a monotonic fashion when 'x' was changed from 0 to 1. The H_c increased by seven times from $x = 0$ to $x = 0.4$, and subsequently decreased by fourteen times from $x = 0.4$ to x = 1.0. The T_c of these alloys also increased monotonically with increasing Co content. In addition to the main phase transition, a second phase transition in a composition range between $x = 0.6$ and $x = 0.8$ was also observed. The T_c for $x = 1$ and $x = 0$ was found to be more than 973 K and ~363 K, respectively. Such substantial changes in M_s, H_c and T_c by changing the composition can be utilized to design these alloys for specific applications.

$Al_xCoCrFeNi$ is a widely studied HEA which can be processed using AM [207,208,222,223]. The magnetic properties of conventionally processed $Al_xCoCrFeNi$ have been studied [188,224–227]. However, there are very few reports of the magnetic properties of AM processed $Al_xCoCrFeNi$ alloys. An alloy library of $Al_xCoCrFeNi$ ($x = 0.15$–1.32) HEAs was prepared by the LENS technique [206]. The structure changed from a single FCC phase to a BCC + B2 two phase structure with increasing Al content (Fig. 21(a)). The hardness increased with Al content within the FCC phase region, while it remained relatively unchanged with increasing Al content in the BCC/B2 region (Fig. 21(b)). The structure and mechanical properties of $Al_xCoCrFeNi$ HEAs prepared by AM and conventional methods were found to be comparable [207,224].

Recently, Karlsson et al. reported that the microstructure of AlCoCrFeNi HEA prepared using the induction melting method and by SLM were quite different [223]. Coarse and fine grains were observed in induction melted alloys and the SLM alloys, respectively. The change in microstructure may be associated with the different cooling rates of the two processes. SLM alloys consisted of the BCC + B2 phases, whereas conventionally cast or deformed alloys exhibited the A2 and FCC phases.

5. Magnetic shape memory alloys

Ni-Mn-Ga magnetic shape memory alloys (MSMA) are multifunctional magnetic alloys, exhibiting both the magnetic shape memory effect [232–235] and the magnetocaloric effect [236–238]. Ni-Mn-Ga MSMA exhibit a magnetic-field-induced strain (MFIS) and have potential applications in energy harvesters, actuators, sensors, delivery micropumps, etc. [235,239–244]. However, the brittle nature and limited MFIS of these polycrystalline Ni-Mn-Ga alloys limit their applications. Numerous processing methods, such as melt spinning [245–247], spark plasma sintering [248,249] and ball milling [235,250,251] have been developed to obtain suitable mechanical and structural properties of these alloys. For example, the MFIS of Ni-Mn-Ga alloys can be improved by increasing porosity and introducing texture [252,253].

Recently, porosity control and improved MFIS of Ni-Mn-Ga MSMA were investigated using AM processed alloys [254–261]. Binder jet 3D printing, followed by sintering at temperatures of up to 1100 °C, was used to produce MSMA with densities of up to 99% [255,261]. Pre-alloyed $Ni_{49.7}Mn_{30}Ga_{20.3}$ powder exhibited the austenite phase, however, twinned 14 M modulated martensite was present in AM processed alloys. Sintering the samples at 1070 °C resulted in a density of 66%, a M_s of 56.5 Am^2/kg and T_c of 90 °C. Ni-Mn-Ga MSMA processed by laser metal deposition showed a 14 M martensitic ferromagnetic structure [254]. The samples exhibited narrow hysteresis and enhanced M_s after homogenizing at 1000 °C for 24 h, followed by ordering at 700 °C for 12 h. Ni-Mn-Ga MSMA sintered at 1000 °C and chemically ordered at 700 °C resulted in porosity levels of 73–76% and a non-modulated martensite phase [260]. A martensite/austenite transformation between 45 and 90 °C, Tc between 85 and 90 °C, and M_s of up to 56 Am^2/kg were

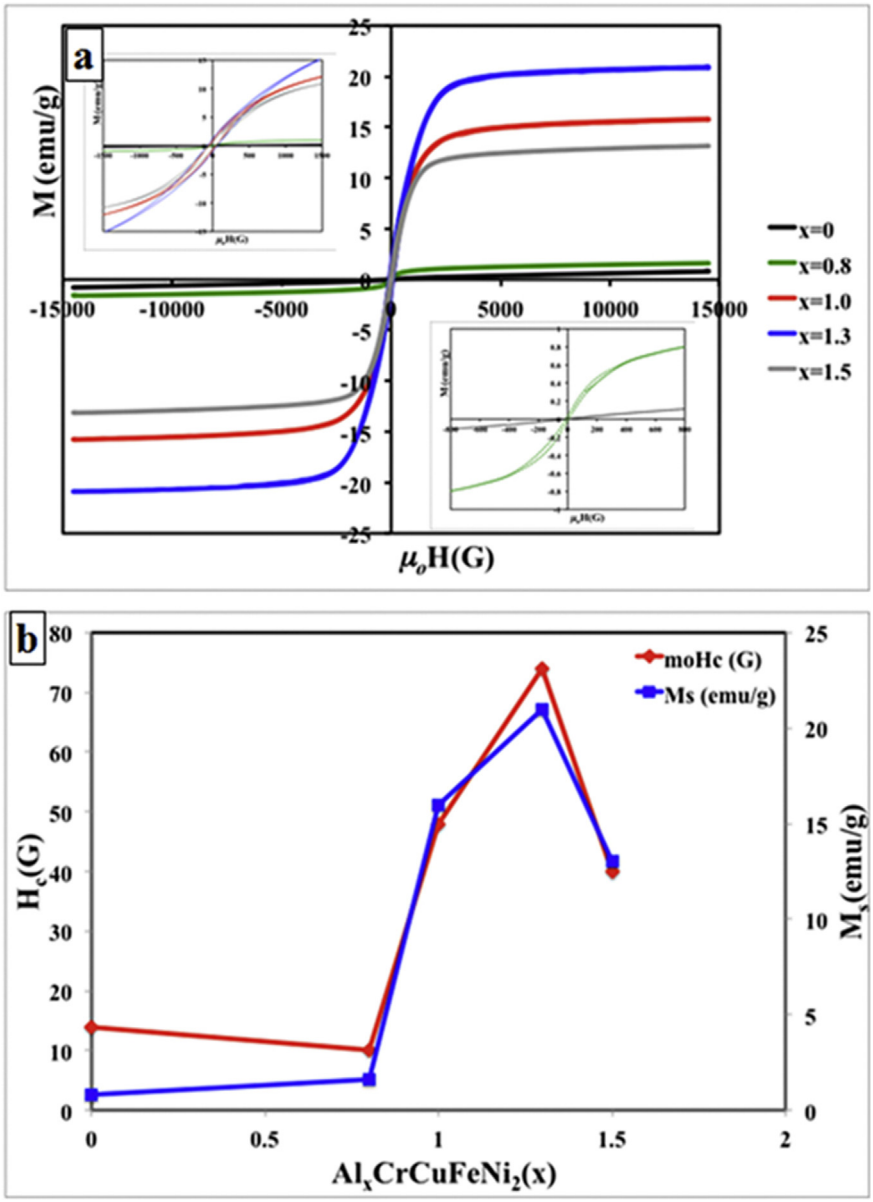


Fig. 19. (a) M-H curves of AlxCuCrFeNi2 alloys for x = 0.0, 0.8, 1.0, 1.3, 1.5, and (b) change in coercivity (H_c) and saturation magnetization (M_s) with Al concentration. Reprinted with permission from [35] © 2017, Elsevier.

achieved. Thus, AM and subsequent heat treatment can introduce porosity to enhance MFIS. The capability to design porous parts with complex geometries by AM can have wide applications in energy management and sensors [252]. The change in magnetic properties with heat treatment condition in summarized in Table 5.

Taylor et al. reported the fabrication of Ni-29Mn-21.5 Ga wires using the extrusion of liquid inks which contain Ni, Mn, and Ga powders [257]. The dependence of average porosity on sintering temperature/time is shown in Fig. 22.

6. Magnetocaloric heat exchanger

Magnetic cooling, based on the magnetocaloric effect, is a green and environmentally benign technology and has attracted considerable interest for thermal management [122]. Magnetocaloric systems require an efficient geometry of heat exchangers made of magnetocaloric materials (MCM). These exchangers should be able to quickly transfer heat to the fluid. Numerous designs of heat exchangers, such as parallel plate arrangements, packed powder beds, tube channels in a solid block, etc. have been proposed [262]. A wavy channel block and an array of fin shaped rods of a $La(Fe,Co,Si)_{13}$ magnetocaloric regenerator have been prepared using SLM (Fig. 23) [263]. Annealing of these structures resulted in ~71% of the $LaFe_{10.7}Co_{1.3}Si$ phase and a reasonable change in magnetic

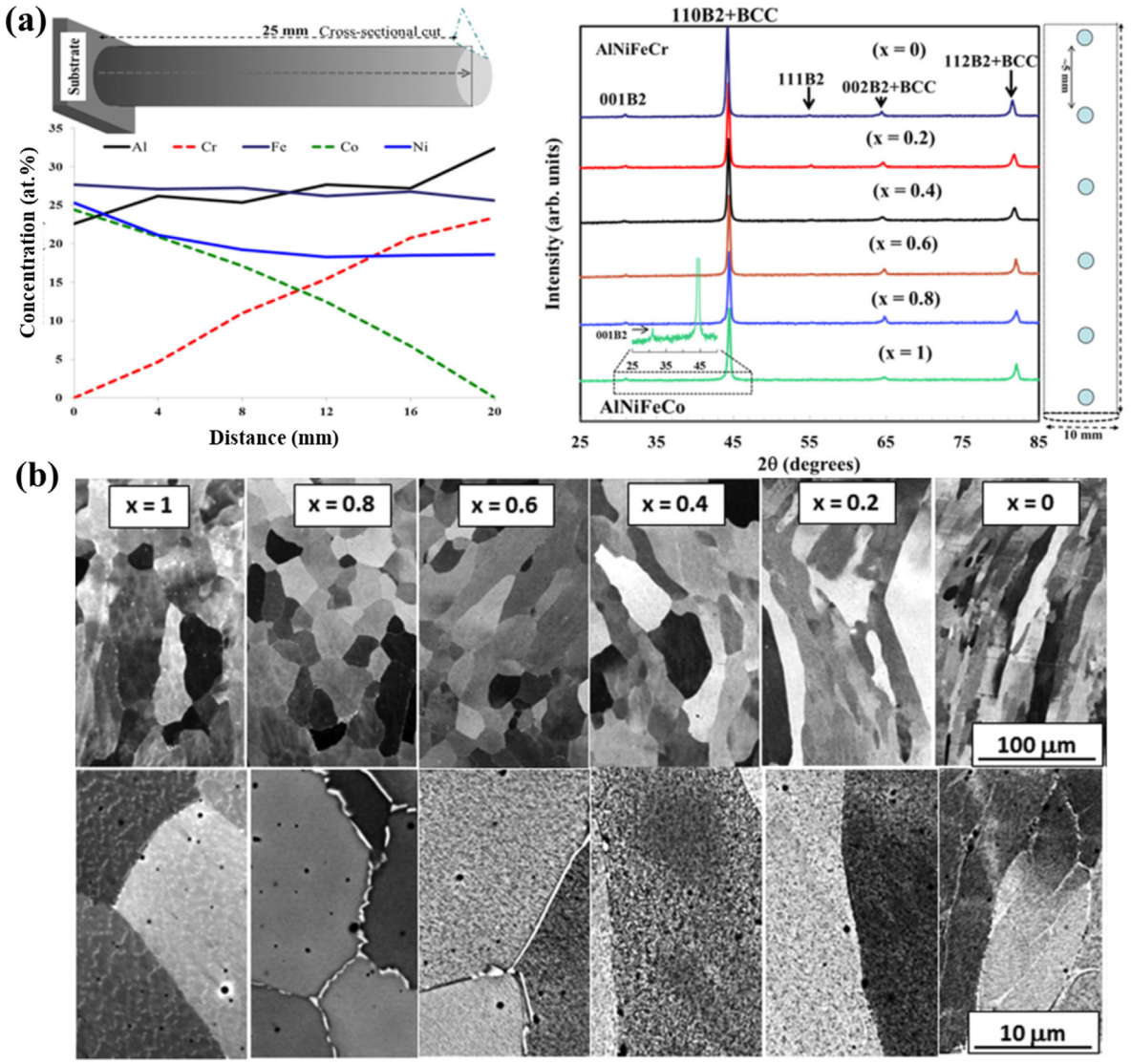


Fig. 20. (a) Composition and XRD data along the height of LENS deposited graded $AlCo_xCr_{1-x}FeNi$ cylinders (b) Backscattered SEM images of the graded $AlCo_xCr_{1-x}FeNi$ alloy. Reprinted with permission from [34] © 2017, John Wiley and Sons.

entropy (ΔS_m) and adiabatic temperature (ΔT_{ad}). The annealed regenerator was stable up to ~10^6 cycles at T_c of 325 K and a magnetic field of 0.6 T (Fig. 23 (e)). The channel diameter of the blocks was 800 μm due to the brittleness and irregular shape of the starting powder. The channel width to achieve optimum operating frequency was found to be ~100 μm [264].

In another study, $La(FeSi)_{13}$-based alloys were fabricated using Laser Beam Melting (LBM) [265]. Geometries of various thicknesses, as well as round and rectangular shaped heat exchangers with channel sizes of 500 μm and 300 μm were prepared. The laser power (P), the scan speed (v), and energy density (E) were varied in the range of 160 W to 190 W, 400 mm/s to 500 mm/s and 25 mJ to 30 mJ, respectively. More cracks appeared in samples produced with high energy density values. Reducing cracks in samples processed by LBM is a challenge, especially for brittle materials, due to the internal stresses developed during rapid cooling [266]. Annealing of LBM processed samples at 1423 K for 2 h showed ~90% of the 1:13 phase with a ΔS_m of -3.8 J·kg^{-1} K^{-1} at a field of 1 T. An $AlFe_2B_2$ cylinder of 5 mm diameter and 30 mm height displayed a ΔS_m of 2.8 J·kg^{-1} K^1 and ΔT_{ad} = 1.1 K at field of 2 T [267]. These MCE values are comparable to those of samples prepared by the suction casting method [268]. $Ni_{43}Co_7Mn_{39}Sn_{11}$ and $Ni_{50}Mn_{18.75}Cu_{6.25}Ga_{25}$ powder was used to prepare various geometries for magnetocaloric applications using the direct laser deposition (DLD) [269,270] and binder jet AM processing techniques [271]. The AM processing parameters can also be used to tune the T_c and magnetocaloric properties.

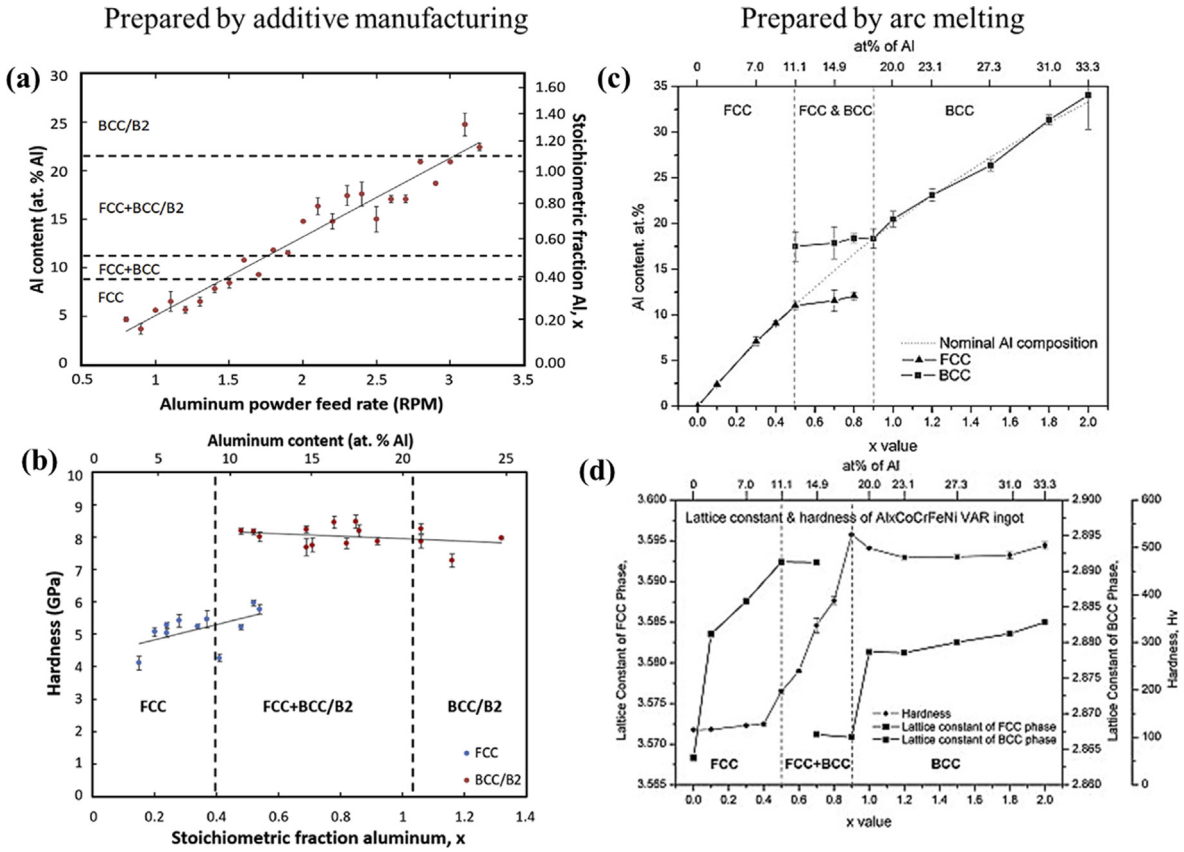


Fig. 21. (a) The phases present in $Al_xCoCrFeNi$ HEA prepared by AM as a function of Al content, (b) Hardness of the FCC and BCC/B2 phases in the $Al_xCoCrFeNi$ alloys prepared by AM, vertical lines indicate the composition ranges where the microstructure is predominantly FCC (low x), multiphase (medium x) and BCC/B2 (high x), Reprinted with permission from [206] © 2018, Elsevier, (c) The phases observed in $Al_xCoCrFeNi$ HEA prepared by arc melting versus Al content and (d) Hardness and lattice constant of the $Al_xCoCrFeNi$ HEAs prepared by arc melting as a function of Al content. Reprinted with permission from [228] © 2012, Elsevier.

Table 5
Magnetic properties of Ni-Mn-Ga magnetic shape memory alloy (MSMA) prepared by AM: saturation magnetization (M_s), coercivity (H_c), Curie temperature (T_c).

Sample	M_s (Am^2/kg)	H_c (mT)	T_c (°C)	Ref.
As deposited	51.8	28	86	[254]
Homogenized at 1000 °C/24 h and ordered at 700 °C/12 h	57	4	90	[254]
Micro-trusses, sintered at 1000 °C/12 h and ordered for 700 °C/10 h	48–56.4	5	85–95	[260]
Sintered at 1020 °C/4h in vacuum	15.3	58	–	[255]
Sintered at 1020 °C/4h in argon	68.4	78	107	[255]

7. Hard magnetic materials

Permanent magnets (PM) are of high interest as they are currently used in a wide range of applications including computers, electric vehicles, electrical machines, refrigerators, washing machines, elevators, TVs, phones, loudspeakers, printers, wind turbines, data storage, etc. Steels, ferrites, Alnico, Sm-Co alloys, Nd-Fe-B alloys are common PM materials.

Sintered Nd-Fe-B alloys, discovered in the early 1980s, have the maximum energy product ($(BH)_{max}$) of ~55 MGOe at room temperature. Besides the magnetic properties, there are other factors (Fig. 24) such as operating temperature, chemical stability, cost, etc., which can be important to select the magnets.

It is clear from Fig. 24 that high performance hard magnets contain expensive rare earths. The conventional techniques to produce these magnets do not efficiently utilize the raw materials. AM has the potential to utilize these materials with minimum waste.

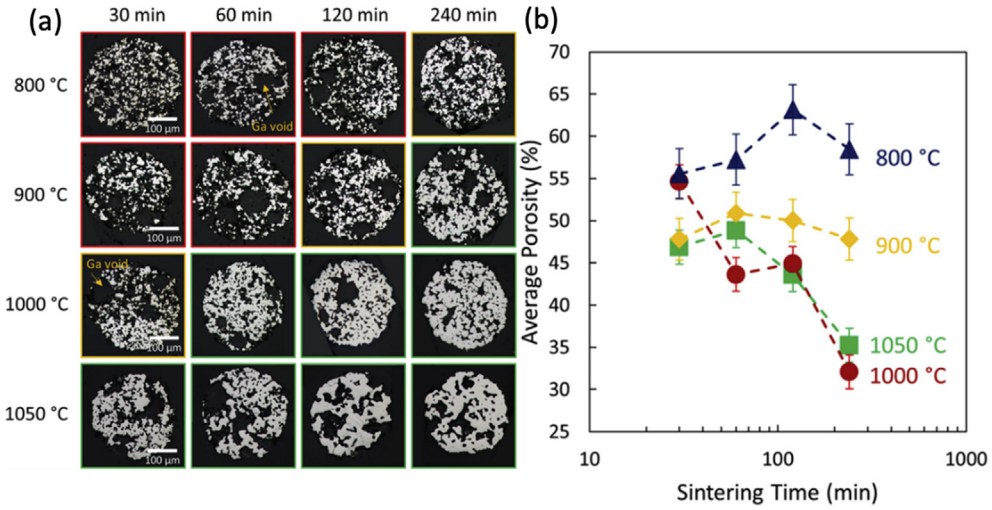


Fig. 22. Optical micrographs of radial cross-sections of Ni-Mn-Ga wires after debinding and sintering at 800–1050 °C for 30–240 min. (b) Average porosity versus sintering time for the printed Ni-Mn-Ga wires shown in (a). Reprinted with permission from [257] © 2013, Elsevier.

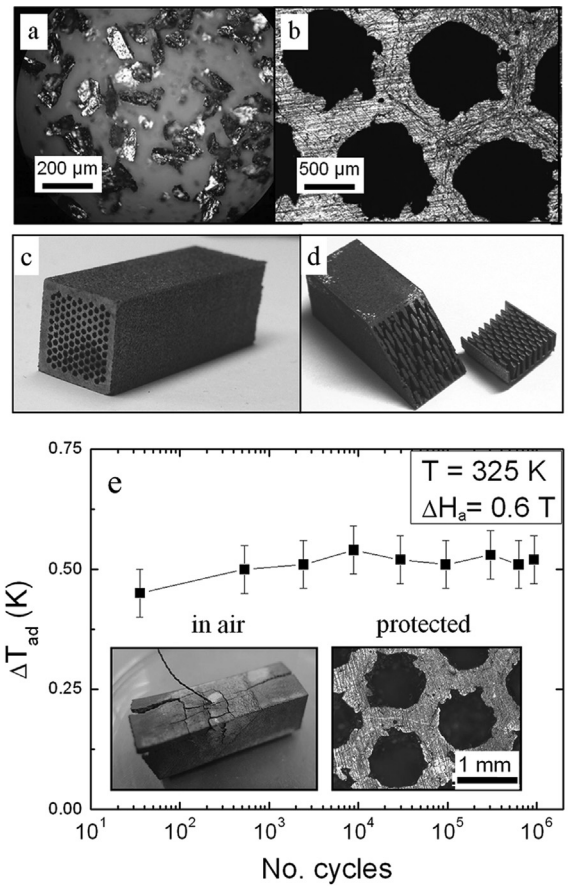


Fig. 23. (a) $LaFe_{11.7}Co_{1.3}Si$ powder used for printing (b) magnified view of wavy channels made by $LaFe_{11.7}Co_{1.3}Si$ using SLM. Blocks of (c) wavy channel and (d) fin shaped rods. (e) Adiabatic temperature change (ΔT_{ad}) versus number of cycles. The insets display cracking of the annealed block. Reprinted with permission from [263] © 2013, AIP Publishing.

7.1. Nd-Fe-B permanent magnets

Nd-Fe-B based magnets can be divided into sintered and bonded magnets. For net-shape production of these magnets, a

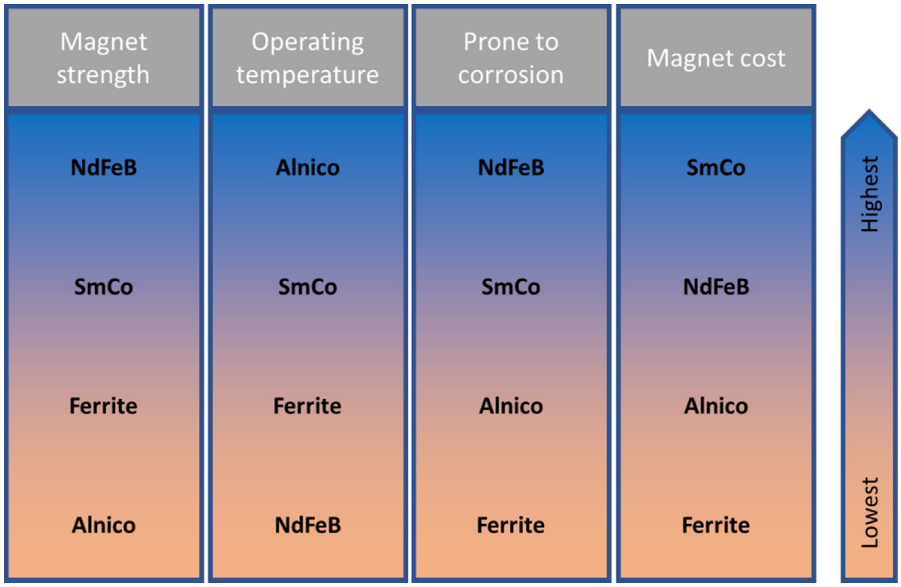


Fig. 24. Selection matrix of permanent magnets.

compromise between complex shape and magnetic properties is a current challenge. Simple geometries can be readily formed but complex shapes can only be produced by adding nonmagnetic binders, reducing the magnetic properties. Isotropic Nd-Fe-B bonded magnets with complex shapes have been printed using big area additive manufacturing (BAAM), direct write and binder jetting techniques.

7.1.1. Bonded Nd-Fe-B magnets

The traditional way to manufacture bonded Nd-Fe-B magnets is to mix magnetic powder in a polymer binder followed by injection molding, compression molding, extrusion, etc. [272]. The volume fraction of magnetic powder in injection molding is ~65% while in compression molding it can be up to ~80%[272].

The fabrication of bonded Nd-Fe-B magnets using AM is of high interest [273–283]. A 3D-printable epoxy-based ink containing anisotropic Nd-Fe-B magnet particles was formulated for direct-write AM [282]. The feedstock contains up to 40 vol% of such particles. Ring, bar, and horseshoe-type magnets were built and cured in air. Huber et al fabricated polymer bonded Nd-Fe-B magnets with a loading of 54% of PM using a 3D printer which uses extruded filaments as the feedstock material [279]. Commercially available isotropic Nd-Fe-B powder inside a polyamide matrix was used for the printing process. Fig. 25 shows the magnetic hysteresis in each direction of magnetization for the samples prepared by the injection molding and 3D printing. The measured remanence and coercivity for 3D printed magnets were found to be 310 mT and 740 kA/m, respectively, while the magnetic properties of injection molded magnets were 387 mT and 771 kA/m, respectively. A complex shape 3D printed magnet has also been designed to generate a specific stray field distribution [284].

Big Area Additive Manufacturing system (BAAM) can produce meter size parts at a deposition rate of up to 50 kg/h using

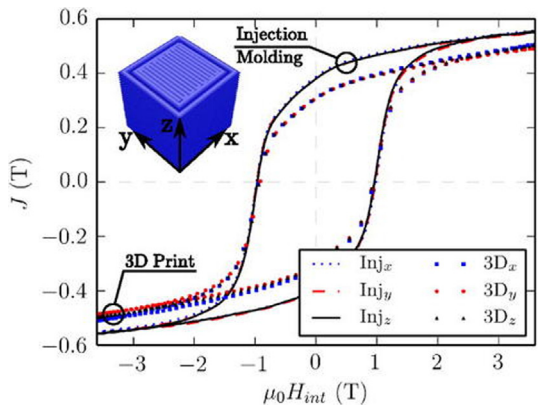


Fig. 25. Hysteresis loops of injection molded and 3D printed magnet measured in different magnetization directions. Reprinted with permission from [279] © 2016, AIP Publishing.

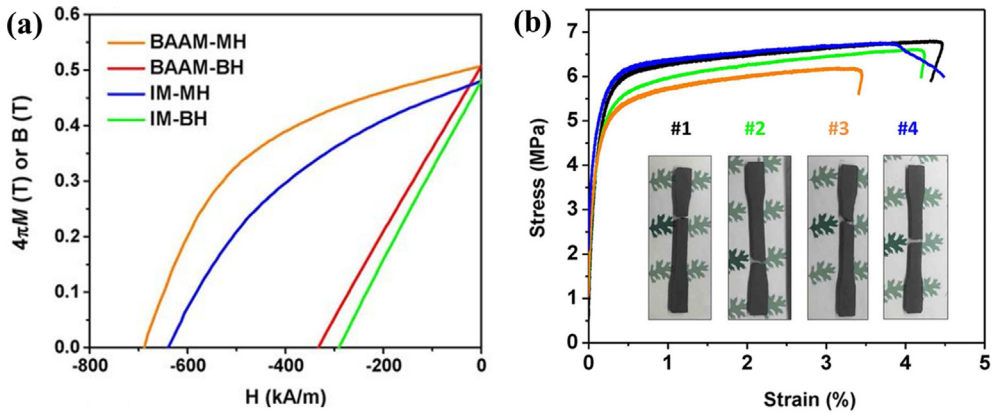


Fig. 26. (a) MH and BH demagnetizing curves for BAAM and injection molded magnets (b) Tensile stress–strain curves of the Nd-Fe-B magnets fabricated by BAAM; the inset shows the location of failure after tensile testing. Reprinted with permission from [274] © 2016, Nature Publishing Group.

feedstock materials such as compounded pellets [285]. Li et al fabricated Nd-Fe-B bonded magnets using BAAM with better magnetic and mechanical properties than those of prepared by injection molding (Fig. 26) [274]. The starting pellets with 35 vol% of polyamide and 65 vol% of isotropic Nd-Fe-B powder resulted the magnet with density of ~4.8 g/cm^3. The intrinsic coercivity (H_{ci}), remanence (B_r) and energy product ((BH)$_{max}$) of the final product were found to be 688.4 kA/m, 0.51 T and 43.49 kJ/m^3 (5.47 MGOe), respectively. This magnet showed a tensile strength of 6.60 MPa and a failure strain of 4.18%. In a follow up study, Nd-Fe-B bonded magnets with a high loading fraction of 70 vol% Nd-Fe-B were fabricated [278]. The density of the printed magnet was enhanced to ~ 5.2 g/cm^3. The H_{ci}, B_r, and (BH)$_{max}$ were found to be 8.9 kOe (708.2 kA/m), 5.8 kG (0.58 T) and 7.3 MGOe (58.1 kJ/m^3), respectively. The tensile strength was found to increase for larger loading fraction of the magnet powder.

Isotropic-bonded magnets were built by powder bed fusion using a feedstock composed of polymeric binder polyamide (PA-12) and a $Nd_2Fe_{14}B$ atomized nanocrystalline powder [286]. A feedstock based on 34% volumetric fraction of PA-12 resulted in a density of 3.6 g/cm^3 and remanence of 0.3 T. There was no significant change in coercivity by changing process parameters, and the coercivity values of ~700 kA/m were retained.

Near-net-shape isotropic $Nd_2Fe_{14}B$-based bonded magnets prepared a binder jetting AM process must compete with conventional injection molding. The density of the binder jet magnets was 3.47 g/cm^3 which is ~46% of the NdFeB single crystal density (7.6 g/cm^3) [273]. Binder jetting AM technique was employed to fabricate Nd-Fe-B isotropic bonded magnets, followed by an infiltration process with eutectic alloys [275]. Densification, mechanical strength improvement, and an enhancement in intrinsic coercivity from 732 to ~1300 kA/m were achieved. Low-melting point alloy diffusion is an effective technique to enhance coercivity and mechanical strength of binder jetted bonded magnets.

7.1.2. Hybrid bonded Nd-Fe-B magnets

Nd-Fe-B composite bonded magnets, with the addition of $Sm_2Fe_{17}Nx$ [287–291] or ferrites [292], showed novel magnetic, mechanical, thermal and corrosion resistance properties. Hybrid bonded magnets such as NdFeB/SmFeN [277] and NdFeB/$SrFe_{12}O_{19}$ [280] have also been fabricated using AM. Anisotropic NdFeB/SmFeN composite bonded magnets with a 65 vol% fraction of magnetic powder were aligned in a magnetic field strength of 2 T. These magnets exhibited a (BH)$_{max}$ of ~11 MGOe [277]. The magnetic properties (B_r, H_c, (BH)$_{max}$) of hybrid NdFeB/$SrFe_{12}O_{19}$ bonded magnets were found to lower than those of Nd-Fe-B bonded magnets [280]. However, they exhibited higher thermal stability; the coercivity temperature coefficient (β) increased from −0.4%/°C to −0.17%/°C with increasing $SrFe_{12}O_{19}$ content from 0 wt% to 25 wt%.

7.1.3. Sintered Nd-Fe-B magnets

To obtain hard magnetic properties, the microstructure of Nd-Fe-B magnet must a have hard magnetic phase of $Nd_2Fe_{14}B$ surrounded by a soft magnetic phase of Nd. Therefore, tuning of laser parameters such as speed, focus, power as well as material layer thickness is vital to obtain a dense net shape hard magnet. Nd-Fe-B magnets have been fabricated using laser based AM techniques [293–298]. SLM of the powders resulted in very fine microstructures with directional crystal growth and a finely dispersed Nd-rich phase [297]. Fine and textured microstructures are helpful to obtain a large value of H_c in sintered Nd-Fe-B magnets. SLM processed magnet and book mold cast (BMC) magnets were used to make sintered magnets. SLM sintered magnet exhibited high coercivity and lower permeability compared to BMC sintered magnets (Fig. 27).

In another study, a comparison between the magnetic performances of 3D printed, injection molded, and spark plasma sintered magnets produced from $Nd_{7.5}Pr_{0.7}Zr_{2.6}Ti_{2.5}Co_{2.5}Fe_{75}B_{8.8}$ powder was performed (Fig. 28) [296]. At room temperature, the B_r and (BH)$_{max}$ values of direct metal 3D printed magnets were higher than those of injection molded and SPS sintered magnets. At elevated temperatures, the (BH)$_{max}$ value of SLM processed magnets was found to be similar to those of injection molded magnets and significantly larger than that of SPS magnets (Fig. 28 (b)).

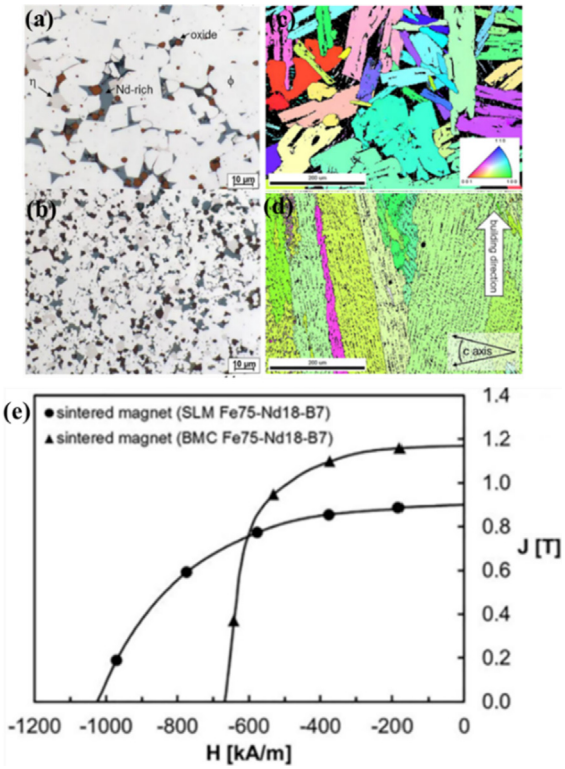


Fig. 27. Microstructure of a sintered magnet made of (a) book-mold-cast (BMC) Nd-Fe-B and (b) SLM Nd-Fe-B. EBSD analysis of (c) BMC Nd-Fe-B and (d) SLM Nd-Fe-B. Hysteresis loops in the second quadrant showing comparison between SLM and BMC processed sintered magnets. Reprinted with permission from [297] © 2019, John Wiley and Sons.

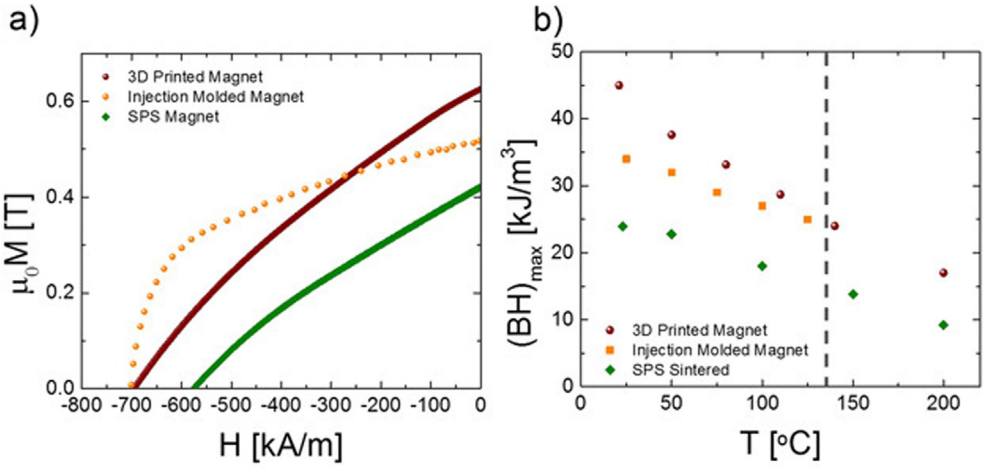


Fig. 28. (a) Hysteresis loops for samples prepared by three different techniques, b) Temperature dependence of $(BH)_{max}$ for these samples. Reprinted with permission from [296] © 2017, John Wiley and Sons.

A grain boundary diffusion (GBD) process was used to enhance the coercivity of NdFeB magnets. Huber et al. applied the GBD process to selective laser sintered NdFeB magnets and demonstrated significantly enhanced values of coercivity [299]. The H_c of the as printed sample was ~517 kA/m which increased to ~795 kA/m after the GBD of Nd-Cu and Nd-Al-Ni-Cu alloys. The coercivity was increased to ~1194 kA/m with GBD of Nd-Tb-Cu. Following the eutectic diffusion process, the samples displayed a full density of ~7.5 g/cm^3. Fig. 29 shows the relevant micrographs, elemental maps and enhanced coercivity of diffusion processed samples.

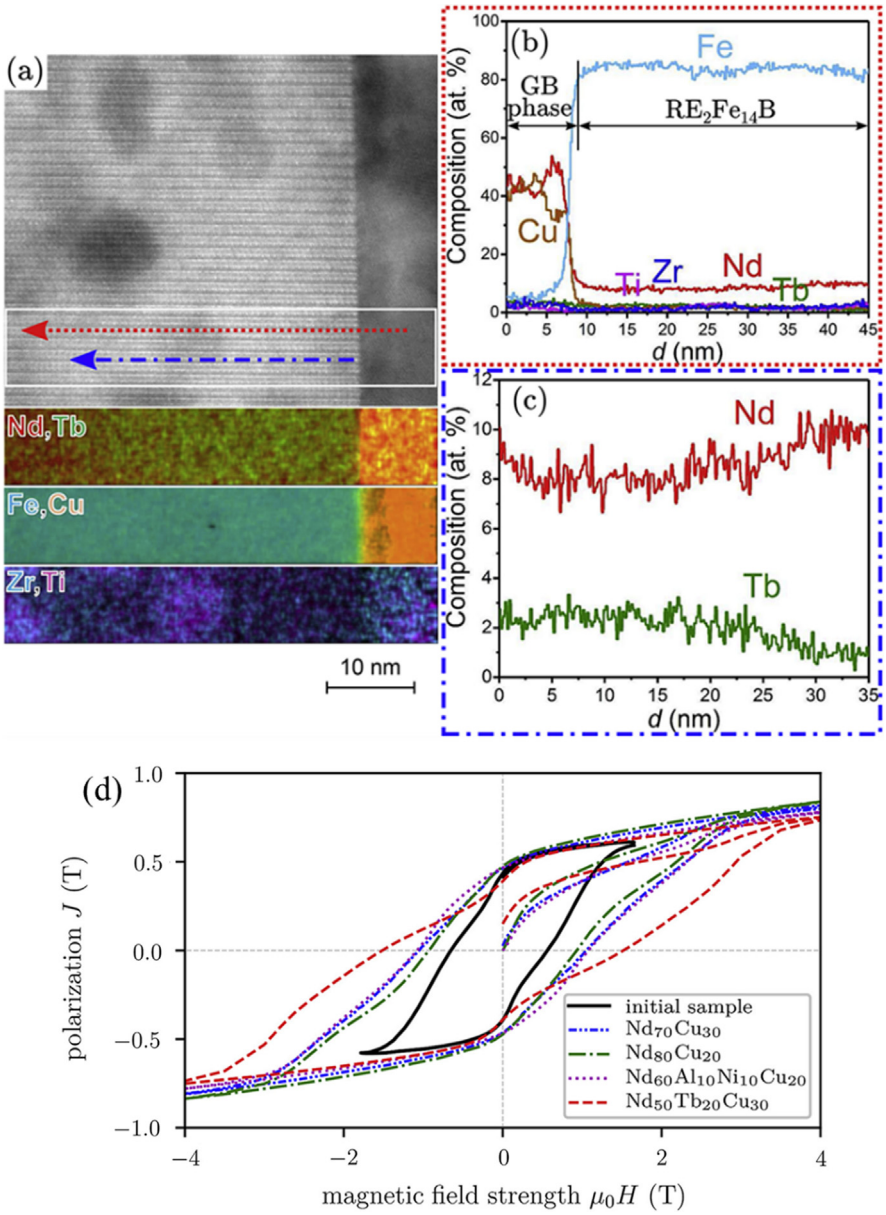


Fig. 29. (a) STEM-HAADF image and STEM-EDS elemental mapping images for GBD processed samples. Line scan of the composition along the (b) red arrow and (c) blue arrow. (d) Hysteresis loops for the initial AM sample and for the diffusion-processed samples. Reprinted with permission from [299] © 2019, Elsevier.

7.2. Alnico permanent magnets

Alnico PM are rare earth free magnets exhibiting moderate coercivity, high remanence, and relatively high-energy product. There is not much literature on the AM of Alnico. Alnico magnets made via using a LENS system has been reported [300]. These magnets showed remanence values of up to 9 kG, which is close to the 10.6 kG value obtained for anisotropic cast alnico 9. The LENS built samples displayed H_c of ~161 kA/m which matched well with the values for sintered alnico 8H. AM processing is a promising method to obtain high coercivity and high remanence alnico permanent magnets, but further research is required in this area.

7.3. Sm-Co permanent magnets

Sm-Co is one of the most expensive magnets which is generally utilized for high temperature applications. One of the drawbacks of these magnets is that they are very brittle. However, fabrication by AM may reduce brittleness. There is very little literature on Sm-Co magnets prepared by AM. This may be because of the limited market and high cost of cobalt and samarium. In addition, formation

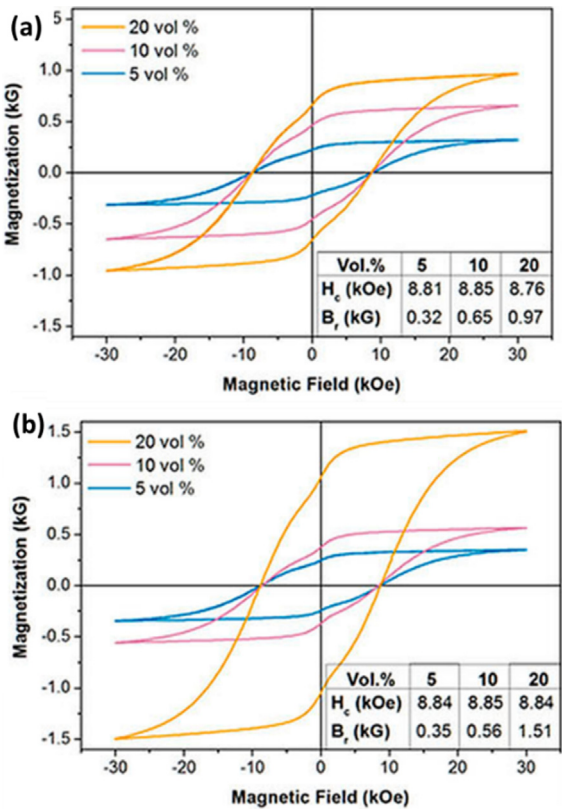


Fig. 30. Magnetic hysteresis of a) un-extruded Sm-Co powder loading in PLA and b) extruded Sm-Co powder loading in PLA. Reprinted with permission from [301] © 2018, AIP Publishing.

of the required phase, which exhibit hard magnetic properties is challenging since Sm-Co alloys can exhibit phases with soft magnetic properties. Khazdozian et al., fabricated bonded magnet filaments using extrusion of recycled Sm-Co powder [301]. A maximum of ~20 vol% of the recycled Sm-Co was blended into polylactic acid (PLA) to produce a filament. The magnetic properties of un-extruded and extruded Sm-Co powder loaded in PLA is shown in Fig. 30 [301].

7.4. Ce-Co-Fe-Cu and mischmetal–Co-Fe–B permanent magnets

Ott et al. used LENS for rapid assessment of Ce-Co-Fe-Cu and MM-Co-Fe-B (MM = mischmetal; La, Ce, Nd, and Pr) alloys for permanent magnet applications [302,303]. Rod shaped samples of with different compositions were printed on a stainless-steel substrate. Fig. 31 shows contour plots of the compositions, T_c and magnetization at 3 T of the rod shape MM-Co-Fe-B samples prepared by LENS. T_c was first found to be increased with increasing the Co fraction upto ~0.23, and then it decreased with increasing Co fraction. Replacing Fe with Co in $RE_2(Fe\text{-}Co)_{14}B$ (RE = Rare earths) systems is known to increase the thermal stability and therefore can be used for high temperature applications [304–307]. The data for magnetization were extracted from hysteresis loops while T_c was measured using magnetic field assisted thermogravimetric analysis [302,303]. This rapid assessment approach is helpful to identify the best combination of magnetic properties and thermal stability.

8. Prototyping of additive manufactured magnetic materials

Different components of electrical machines including magnetic cores and rotors have been developed using AM technologies [308–312]. The transformer, and stator for electric machines have been prepared using a 3d-multimaterial printing process (Fig. 32) [313]. In this process, highly viscous pastes were used to build a three-dimensional specimen in layers. The starting materials of the pastes may be powders of metals, alloys or ceramics. During the sintering process of the 3d printed parts, the binder was expelled, and the metallic/ceramic particles diffused together. A light weight rotor was fabricated using SLM and used to make the prototype [314]. The weight was reduced by implemented cavity and lattice structures into the rotor while maintaining the strength. Gas atomized Fe-Co powder with particle size not more than 63 μm was used to build a specially designed rotor using SLM [311]. A density of ~99.9% can be achieved by optimizing the process parameters.

Rotor shafts manufactured by AM were assembled into a conventionally laminated stator, and the functionality of the machine was tested [315]. Different 3D printed geometries were tested using an integrated magnetic circuit. An effective 3D printed

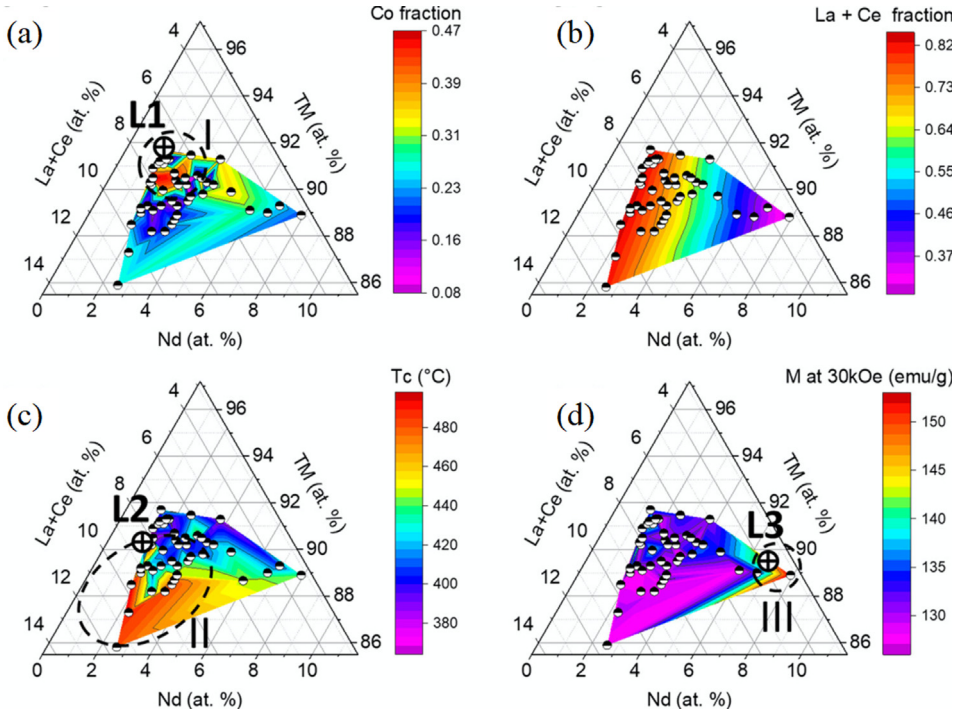


Fig. 31. Contour plots of LENS processed MM-Co-Fe-B (a) Co fraction in Co-Fe (TM), (b) La + Ce fraction in MM, Compositional dependence of (c) Curie temperature and (d) magnetization at 30 kOe. Reprinted with permission from [303] © 2020, American Chemical Society.

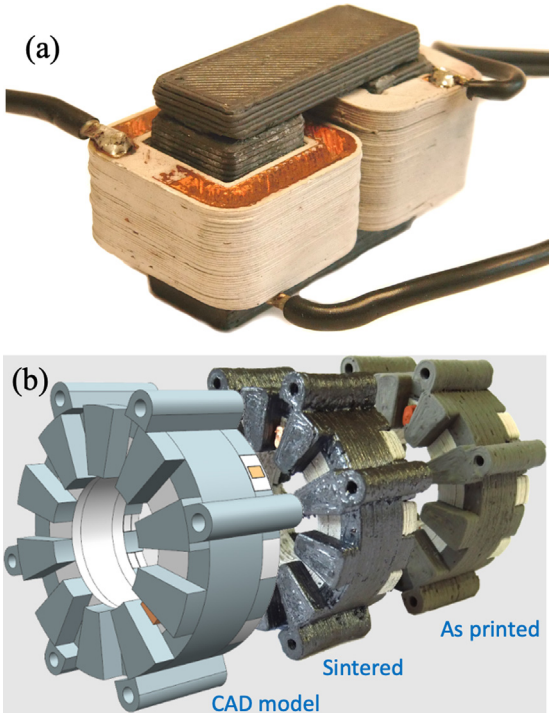


Fig. 32. (a) Transformer manufactured by 3D multi-material printing process, (b) Stator of a winding-less reluctance machine, Reprinted from [313] © Chemnitz University of Technology.

transformer core need to have a high fill factor and high volume fraction of magnetic particles [316]. An electric motor with hybrid-field topology was introduced. 3D magnetic flux paths are allowed by isotropic soft magnetic material built by a AM technique based on spray forming technology [317,318]. The motors provide a high power output and improved energy efficiency compared to the

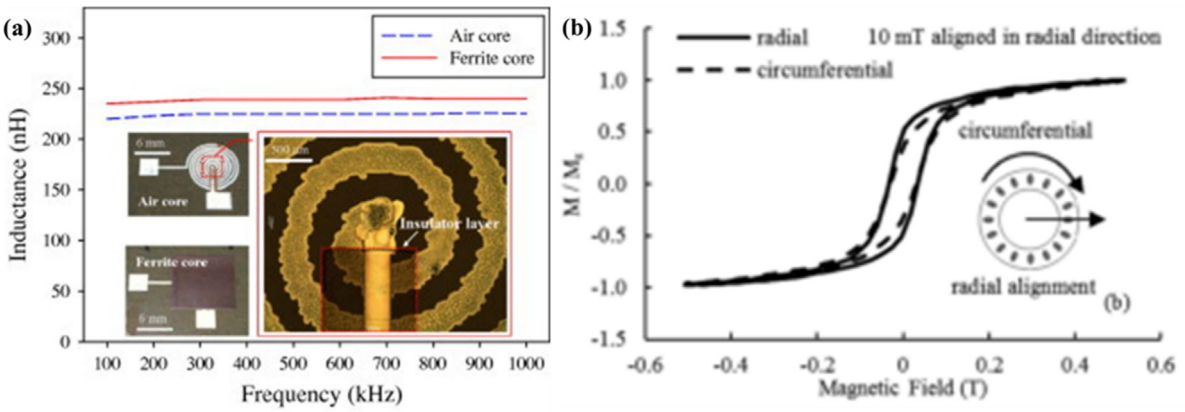


Fig. 33. (a) Printed flat spiral inductor with air and ferrite core, and their inductance at frequency up to 1 MHz. Reprinted with permission from [319] © 2012, Elsevier. (b) Hysteresis curves measured in the radial and circumferential directions for aligned ring-shaped sample. Reprinted from [56] © 2014, Oregon State University.

conventional systems [317].

Fig. 33(a) shows the inductance of an inductor printed by ink-jet printing as a function of applied frequency [319]. The inductor with printed ferrite core exhibited high inductance with almost no change up to 1 MHz. Dhagat et al. reported 3D printed Co-based magnetic materials with arbitrary anisotropy [56,320]. Fig. 33(b) shows a more square hysteresis response in the radial direction than in the circumferential direction [320]. Therefore, the performance of 3D printed magnetic devices can be tuned according to the specific application by controlling magnetic anisotropy. Inkjet printing has the potential to be utilized in various electronic components such as magnetic sensors, antenna, etc.

9. Challenges associated with AM prepared magnetic materials and components

Since AM is a relatively new manufacturing technology and not yet fully matured, there are several variables which need to be controlled, monitored, and optimized during the AM process for the best performance of the 3D printed part. Most of the effort has been focused on optimization to obtain superior magnetic properties of AM processed magnetic materials. Soft magnetic properties alone may be insufficient for some rotating electrical machine applications, with high strength, low electrical loss and high resistivity also being important parameters. Experimental efforts to produce AM parts can be minimized by machine learning (ML) and artificial intelligence (AI) assisted AM. AI and ML assisted AM equipment linked with high throughput synthesis, rapid automatic characterization tools, and appropriate robotics will be promising for real time build control and optimization of novel materials processed using these technologies. The Alliance for the Development of Additive Processing Technologies (ADAPT) is using the ML approach to map variables and outcomes for metal AM. General Electric has also suggested the use of ML in AM [321].

Innovative electrical machine (EM) design is of high interest since emerging electrical transportation systems have led the search for alternative production methods of EM. Due to the ubiquitous nature of EM, even a small increase in EM efficiency would have a significant impact on energy consumption and reduction of carbon emissions. AM enables tool free production of components directly from a CAD file. AM technology is not currently used in EM industrial production, but interest in this area is growing rapidly. The majority of research so far has focused on the development of magnetic materials and building components such rotors, stators, copper windings using AM. The manufacturing of fully functional electric machines which contain magnetic and electronic components along with other components such as shafts, bearings, frames, cooling fans, wires, etc., using AM is yet to be achieved [322,323]. Although AM is very versatile, there is a big difference between making single components and manufacturing a complete machine which includes diverse components with a variety of compositions and properties. The surface finish of AM components especially components with small feature sizes is another challenge. Hence building a high performance electric motor using AM requires a combined effort of electrical engineers, mechanical engineers and materials scientists. The conventional method of fabrication and assembly of the component of electric machines makes ambitious design changes difficult. AM can easily be helpful in rapidly developing prototypes of novel designs, facilitating the development of high performance next generation electrical machines.

Soft magnetic composites (SMC) exhibit low iron losses at high-frequencies [324]. Additionally, the 3D structure of SMC can reduce the weight and copper losses compared to electrical steel (ES) lamination. However, low permeability, low output torque and low mechanical strength are some of the main drawbacks of SMC cores due to the nonmagnetic binder content. Recently, AM has been used to fabricate a hybrid stator utilizing both 2D ES sheet laminations and 3D end cap SMC structures [325]. Motors with hybrid stators showed high permeability and lower iron and copper losses. The hybrid stator core had ~12% less weight compared to a conventional ES stator core. The efficiency of the hybrid stator was found to lie between that of SMC and ES cores. Several other hybrid approaches have been used to prototype electrical machines, such as inserting components during printing or assembling them after production [326]. More research on the electromagnetic field distribution, functional integrity of cooling effects, core loses, is required to establish AM in the EM industry. AM as a developing technology can produce customized, heterogeneous,

compositionally and functionally graded components. Hence the future potential of AM in the production of magnetic components is bright.

10. Summary and outlook

The fabrication of a variety of magnetic materials using various AM techniques has been reviewed in this paper. The ongoing activities in research groups around the world indicate that AM using a feedstock consisting of either blended elemental powders, or pre-alloyed powders, or metallic powders blended in a suitable polymer, is a viable manufacturing process to develop magnetic parts. AM can produce near net-shape magnets with improved magnetic properties and reduced processing costs. Many soft magnetic materials such as Fe-Ni, Fe-Co, Fe-Si, ferrites, magnetic composites, and high entropy alloys have been successfully fabricated using AM, especially using SLM and LENS type processes. The effect of process parameters such as laser speed, laser power, and heat treatment on the microstructure, mechanical properties, and magnetic properties has been discussed. AM processed bonded Nd-Fe-B permanent magnets exhibit better or equivalent magnetic properties compared to conventionally processed magnets. Two AM methods; i.e., binder jetting and big area additive manufacturing have been successfully used to print such bonded magnets. AM machines equipped with controlled magnetizing tools may be useful to print anisotropic hard magnets. Thus, there is a bright future for additive manufacturing of magnetic materials and components. Additionally, AM will very likely enable the implementation of novel and radical design concepts for magnetic components, such as functionally graded components with tailored site-specific magnetic properties. The future of AM technology will include integrated systems coupling (i) thermal sensors which can provide in-situ diagnostics during printing, (ii) temperature controlled building platform which can control the thermal gradient during printing, (iii) robotic arms for removing the support materials and surface finishing, (iv) high temperature furnace which can be used for post printing heat treatments an automated manner, and most importantly and (v) an improved interface between hardware and software to control every step of the 3D printing process. Such an integrated systems can substantially accelerate AM usage for magnetic material processing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Dr. Varun Chaudhary is a Postdoctoral Research Fellow at the School of Materials Science and Engineering in Nanyang Technological University (NTU) Singapore. He received his PhD from the Interdisciplinary Graduate School, NTU Singapore, his M.Tech. in Materials Science from Indian Institute of Technology Kanpur, India, and his M.Sc. in Physics from Chaudhary Charan Singh University, Meerut, India. He was a DST Research Fellow in National Physical Laboratory, New Delhi, and Research Scientist in Rolls-Royce Corporate Lab, Singapore. He is an editorial board member of Scientific Reports (Nature publishing group) and a member of MRS USA, IEEE and MRS Singapore.

Dr. Chaudhary has strong research expertise in a variety of magnetic materials for energy applications. His research focuses on physical and chemical synthesis, advanced characterization and property evaluation of magnetocaloric materials, soft magnets, permanent magnets, high entropy alloys and functionally graded materials. His current research focuses on accelerated magnetic materials development using high throughput approaches such as additive manufacturing, artificial intelligence and machine learning.



Dr. Srinivas Aditya Mantri received his PhD in Materials Science and Engineering from University of North Texas. After finishing his post-doctorate, he is now working as an Instructional Lab Manager in the Center for Agile and Adaptive Additive Manufacturing at University of North Texas. Dr Mantri has published more than 30 peer reviewed papers in esteemed journals. His area of focus is in additive manufacturing, titanium alloys, and electron microscopy, and deformation behavior of metals. In his spare time, he enjoys visiting state and national parks, and climbing rocks.



Raju V. Ramanujan is Full Professor, School of Materials Science and Engineering, Nanyang Technological Univ., Singapore. He earned his undergraduate and Ph.D. degrees from IIT-Bombay and Carnegie Mellon Univ. (USA), respectively. He is a Fellow of the American Society for Materials and Senior Member of IEEE. He was elected to the Functional Materials Divisional Council, Magnetic Materials Committee (Chair), Phase Transformations Committee as well as Awards Committees of TMS (USA). He was the chair of the IEEE INTERMAG conference and is an IEEE Technical Committee member. Ramanujan serves/served in an Editorial capacity for *Scientific Reports (Nature Publishing Group), Nanomedicine, Metall. and Mater. Trans., Mater. Science and Engineering B*, and *Mater. Science and Engineering C*. He received the Nanyang Award for Excellence in Teaching and the Rolls Royce Inventors award. He is a Visiting Professor at South China Univ. of Tech., Univ. of N. Texas, Indian Institute of Technology and Mumbai University. He has served as Assistant Chair of his School. His research interests include accelerated materials development of functional and structural materials as well as magnetic nanotechnology for sustainability, energy, bio X and robotics applications.



Dr. Raj Banerjee is a University Presidential Professor and Regents Professor in the department of materials science and engineering at the University of North Texas (UNT). His primary research focus is on advanced metallic and functionally graded composite materials for aerospace, energy, and biomedical applications. The use of advanced characterization techniques, spanning over a range of length scales, including scanning and transmission electron microscopy and atom probe tomography (APT), constitute a common thread tying his multiple research activities. Additionally, he focuses on processing of graded alloys, as well as in situ composites, using additive manufacturing techniques such as laser engineered net shaping (LENS). He has over 250 publications in peer-reviewed journals and an H-index of 56 (Google Scholar).